FRM 核心考点速通笔记

FOCO 备考速通系列

说明

FRM 考试的主要参考题目是 3 套 PE 以及机经回忆，基于这些题目，我们整理了相应的考点和保姆级的习题详解，希望这份笔记能够帮助大家快速掌握 FRM 考试的核心考点。这份笔记特点如下：

（1）在教材整体框架下梳理考点

（2）覆盖 PE1、PE2 和 PE3 的 300 道题的考点

（3）同时也覆盖近两年的考经考点

（4）每个习题都有详细的解析，尽可能做到保姆级地讲解

（5）高度浓缩，全是重点

（6）单栏排版，80g 护眼纸

（7）附有考点目录和 PE 习题索引目录，查找方便

（8）可以补充相关要点在上面，形成自己的备考笔记

FRM 备考建议：

1. 聚焦高质量的习题，并结合考经和机经进行复习
2. 注重知识点的理解，而不是死记硬背

考点目录

Liquidity and Treasury Risk Measurement and Management
流动性和财库风险计量和管理

1 Liquidity Risk And Leverage

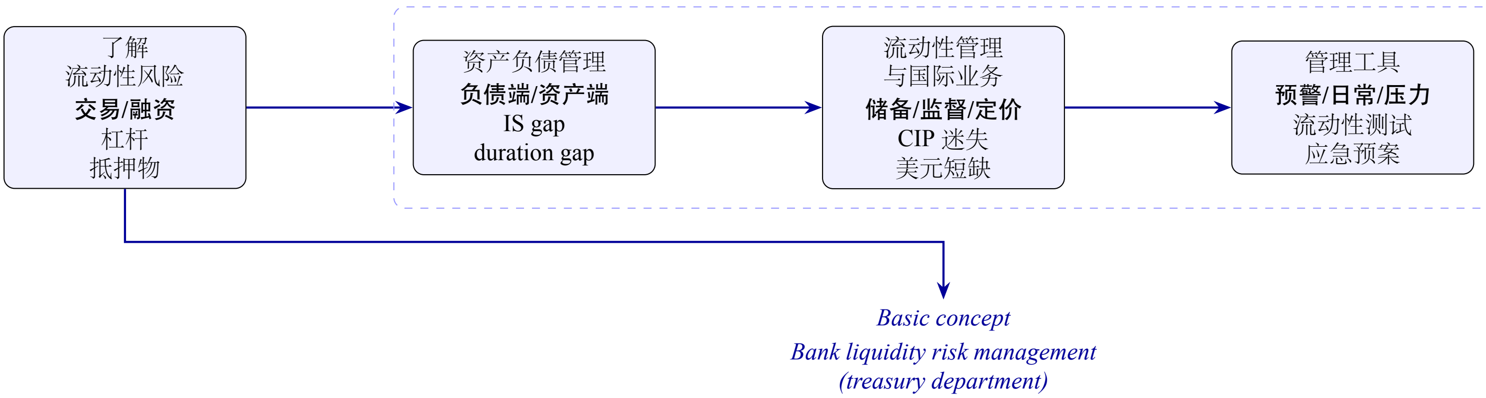


fig 1: 银行流动性风险管理简明流程图

考点 1 Introduction to liquidity risk

- Solvency vs. liquidity
  - Solvency: assets more than liabilities (equity is positive).
  - Liquidity: ability of a company to make cash payments as they become due.
- Two type of liquidity
  - Transaction liquidity: an asset is easy to exchange for other assets.
  - Funding liquidity: is the ability to finance assets continuously at an acceptable borrowing rate.

1 Sources of liquidity risk

- **Transaction liquidity risk(a.k.a. Market liquidity risk or trading liquidity risk):** the risk of moving the price of an asset adversely in the act of buying or selling it.
  - Transaction liquidity risk is low if assets can be liquidated quickly, cheaply, and without moving the price too much.
- **Funding liquidity risk (a.k.a. Balance sheet risk) :** the risk that creditors either withdraw credit or change the terms.
  - Borrower's credit quality is or perceived to be deteriorating and borrower's financial conditions are deteriorating.
- Systemic risk: the risk of a general impairment of the financial system.
  - In situations of severe financial stress, the ability of the financial system to allocate credit, support markets in financial assets, and even administer payments and settle financial transactions may be impaired.
- These types of liquidity risk interact.
  - Funding liquidity risk increases transaction liquidity risk: If collateral requirements or the cost of financing increase, the trader may have to unwind securities.

- Transaction liquidity risk increases funding liquidity risk: If a leveraged market participant have illiquid assets on its books, its funding will be in greater jeopardy.
- Systemic risk increases transaction and funding liquidity risk.

2 Measuring market liquidity

- Financial institutions can sell the asset at the bid to market maker and buy at the offer from market maker.
  - The market maker is likely to increase the bid-offer spread above a certain size.
  - The expected transactions cost is the half-spread or midto-bid spread.
- For asset, measure of market liquidity is its bid-offer spread
  - Dollar amount: Offer price - Bid price.
  - Proportional spread: $s = \dfrac{\text{Offer price} - \text{Bid price}}{\text{Mid market price}}$
- Example
  - Suppose a dealer quoted the bid price and the offer price for a particular bond at 98 and 102 dollars, respectively. Calculate the dollar amount spread(p) and proportional spread(s).
- Answer:
  - Offer price - Bid price = 102 - 98 = 4 dollars
  - Mid price $= \dfrac{\text{offer price} + \text{bid price}}{2} = \dfrac{102+98}{2} = 100$ dollars
  - $S = \dfrac{\text{offer price} - \text{bid price}}{\text{Mid market price}} = \dfrac{102-98}{\dfrac{102+98}{2}} = 4\%$
- For a book, measuring market liquidity is Cost of liquidation
  - In normal market conditions:

$$\text{Cost of liquidation} = \sum_{i=1}^{n} \frac{1}{2} s_i \, P_i \, Q_i = \sum_{i=1}^{n} \frac{1}{2} (\text{offer price}_i - \text{bid price}_i) \, Q_i$$

$$s_i : \text{proportional spread}$$
$$P_i : \text{mid price}$$
$$Q_i : \text{liquidation volume}$$

- Example
  - Tom's portfolio has two bond positions. The related information is shown in the table. What is the cost of liquidation of the whole portfolio in normal market?

Portfolio	Bond A	Bond B
Market value($)	40000	46000
Proportional spread	10%	20%

  - Answer:

$$\begin{aligned}\text{Cost of liquidation of the whole portfolio} &= \sum_{i=1}^{n} \frac{1}{2} s_i P_i Q_i \\ &= \frac{1}{2} \times 10\% \times 40000 + \frac{1}{2} \times 20\% \times 46000 = 6600\end{aligned}$$

- For a book, measuring market liquidity is Cost of liquidation
  - In stressed market conditions:

* Cost of liquidation $= \sum_{i=1}^{n} \frac{1}{2}\left(\mu_i + \lambda\sigma_i\right) P_i Q_i$

$\frac{1}{2}\left(\mu_i + \lambda\sigma_i\right)$: Spread risk factor μ_i : mean of proportional spread

σ_i : standard deviation of proportional spread

λ : critical value under certain confidence level

P_i : mid price

Q_i : liquidation volume

- Example
  - Given the following extra information, what is the 99% confidential interval on the transaction cost per unit of bond A and the 99% spread risk factor of bond B and what is the cost of liquidation of the whole portfolio in stressed market with confidence level of 99%?

Portfolio	Bond A	Bond B
Market value at mid price($)	40000	46000
Shares	400	460
Mean of spread(μ_i)	10%	20%
Standard deviation of spread(σ_i)	6%	8%

  - Answer
    * 99% expected transaction cost per unit of bond A:

$$= \frac{1}{2} \times (10\% + 2.33 \times 6\%) \times (40000 \div 400) \times 1 = 11.99$$

    * 99% spread risk factor of bond B:

$$= \frac{1}{2} \times (20\% + 2.33 \times 8\%) = 0.1932$$

    * Cost of liquidation of the portfolio: $= \sum_{i=1}^{n} \frac{1}{2}(\mu_i + \lambda\sigma_i)P_iQ_i$

$$= \frac{1}{2} \times (10\% + 2.33 \times 6\%) \times 40000 + \frac{1}{2} \times (20\% + 2.33 \times 8\%) \times 46000 = 13683.20$$

- Liquidity-adjusted VaR is regular VaR plus the cost of liquidation.
  - In normal market:

$$\text{Liquidity-adjusted VaR} = VaR + \sum_{i=1}^{n} \frac{1}{2} s_i P_i Q_i$$

  - In stressed market:

$$\text{Liquidity-adjusted VaR} = VaR + \sum_{i=1}^{n} \frac{1}{2}(\mu_i + \lambda\sigma_i)P_iQ_i$$

- Example
  - Allen holds a portfolio with 100,000 shares of a stock and the value of the position is \$4.0 million. The bid price of the stock is \$39 and the offer price is \$41. The stock's daily volatility is 2.59%. Allen is interesting in calculating the liquidity-adjusted value at risk of the portfolio, Let's assume the stock's arithmetic returns are normally distributed (aka, normal VaR) and the expected daily return rounds to zero. Which is NEAREST to the position's one-day 99.0% confident liquidityadjusted value at risk in normal market?

- Answer
  - One day 99.0% VaR: $= |0 - 2.59\% \times 2.326| \times \$4.0\,\text{million} = \$240{,}973.60$
  - The liquidation cost: $= \frac{1}{2} \times \dfrac{41.00 - 39.00}{40.00} \times \$4.0\,\text{million} = \$100{,}000.$
  - The one day 99.0% LVaR: $= 240{,}973.60 + 100{,}000 = \$340{,}973.60$

- VaR translation considering market liquidity:
  - According to previews knowledge.

$$\text{T-day VaR} = \sqrt{T} \times VaR_{\text{1-}day}$$

  - Avoid adverse price impact, the portfolio position equally decreasing every day:

$$\text{Adjusted T-day VaR} = \sqrt{\frac{(1+T)(1+2T)}{6T}} \times VaR_{\text{1-}day}$$

- Example
  - Suppose a one-day VaR of the stock position is 23,000 dollars. To avoid adverse price impact, the fund manager plans to equally liquidate the position in 10 days. What is the adjusted 10-day VaR during the liquidation period?
  - Answer:

$$\begin{aligned}\text{Adjusted 10-day VaR} &= \sqrt{\frac{(1+T)(1+2T)}{6T}} \times VaR_{\text{1-}day} \\ &= \sqrt{\frac{(1+10)(1+2\times 10)}{6\times 10}} \times 23{,}000 = 45{,}129.26 \text{ dollars}\end{aligned}$$

3 Funding liquidity risk

- Funding liquidity risk is the risk of financial institution's ability to cannot finance its cash needs as they arise.
  - The core function of a commercial bank is to take deposits and provide loans. In doing so, it carries out a liquidity and maturity, as well as credit transformation.
  - Maturity mismatch leads to funding liquidity risk
    - To manage funding liquidity risk, bank using asset liability management(ALM).
- Financial institutions that are solvent (i.e., have positive equity) and sometimes fail because of liquidity problems.
  - Northern Rock
  - Ashanti Goldfields
  - Metallgesellschaft

4 Case of Northern Rock

- Case brief: Northern Rock was a fast-growing medium-sized mortgage bank based in the United Kingdom. The bank relied on selling short-term debt instruments for funding.
  Following the subprime crisis, the bank found it difficult to roll over the debt even though the bank was solvent. In November 2011, Northern Rock pie was bought from the British government for £747 million by the Virgin Group.
- Lessons: Liquidity can lead to serious problems even if the bank is solvent.

5 Case of Ashanti Goldfields

- Case brief: Ashanti Goldfields had sought to protect its shareholders from gold price declines by selling gold forward. On September 26, 1999, 15 European central banks announced that they would limit their gold sales over the following 5 years. The price of gold jumped up over 25%.

Ashanti Goldfields was unable to meet margin calls, and this resulted in a major restructuring.

- Lessons: Liquidity problems can arise when hedging illiquid assets with contracts that are subject to margin requirements.

6 Case of Metallgesellschaft

- Case brief: Metallgesellschaft sold a huge volume of 5 to 10 year heating oil and gasoline fixed-price supply contracts to its customers at six to eight cents above market prices. It hedged its exposure with long positions in futures. As it turned out, the price of oil fell and there were margin calls on the futures positions. The outcome was a loss to MG of $1.33 billion.
- Lessons: Liquidity problems can arise when hedging illiquid assets with contracts that are subject to margin requirements

考点 2　liquidity challenges

1 Challenges faced by different institutions

- Commercial banks:
  - Bank run: due to fractional-reserve, if depositors are concerned about banks' liquidity, they will endeavor to redeem before other depositors and lenders.
- Security firms:
  - Lenders abruptly withdrawing credit.
  - Brokerage and clearing customers withdrawing deposits.
- Money Market Mutual Funds (MMMF):
  - Sometimes MMMF had to liquidate assets with loss to meet huge redemption.
    - Break the buck phenomenon.
- Hedge funds:
  - Hedge fund capital that can be redeemed does not only include which will be expected to experience large losses, but also those profitable or having low losses hedge funds.
    - No access to wholesale funding and rely on other tools.

2 Sources of Liquidity

- Holdings of cash and Treasury securities
- The ability to liquidate trading book positions
- The ability to borrow money at short notice
- The ability to offer favorable terms to attract retail and wholesale deposits at short notice
- The ability to securitize assets (such as loans) at short notice
- Borrowings from the central bank

3 Regulations

- Basel III introduced two liquidity risk requirements: the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR).

$$\text{LCR:} \quad \frac{\textit{High quality liquid assets}}{\textit{Net cash outflows in a 30-day period}} \geq 100\%$$

$$\text{NSFR:} \quad \frac{\textit{Available Amount of stable funding}}{\textit{Required amount of stable funding}} \geq 100\%$$

- BIS issued 17 principles for sound liquidity risk management.
  - Identify the responsibilities, procedures, policies, strategies, information disclosure, supervision etc.

4 Liquidity black holes

- Also called ”crowded exit”: a situation where liquidity has dried up in a particular market because everyone wants to do the same type of trade at the same time.
  - During the sell-off, liquidity dries up
    - ∗ When positive feedback traders dominate the trading, market prices are liable to be unstable and the market may become one-sided and illiquid.

5 Positive and negative feedback traders

- Negative feedback traders: buy when prices fall and sell when prices rise.
- Positive feedback traders: sell when prices fall and buy when prices nse.

6 Reasons of positive feedback trading

- Trend trading
- Stop-loss rules
- Dynamic hedging
- Creating options synthetically
- Margins
- Predatory trading
- Relative value fixed income trade

考点 3 Collateral and leverage

1 Collateral market

- Collateral plays an important role in credit transactions by providing security for lenders and thus ensuring the availability of credit to borrowers.
- Transactions in collateral market:
  - Margin loans
  - Repurchase agreements
  - Securities lending
  - Total return swaps
- Risks in collateral market transactions:
  - Market risk: price fluctuation; interest rate fluctuation
  - Credit risk: credit quality of the collateral deteriorates
  - Counterparty risk: counterparty may default

2 Leverage

- **Leverage ratio** $L = \frac{A}{E} = \frac{D+E}{E} = 1 + \frac{D}{E}$
- **Leverage effect** can be seen by writing out the relationship between asset returns r_A, equity returns r_E, and the cost of debt r_D:
  - $\boldsymbol{r_E = r_A + \frac{D}{E} \times (r_A - r_D)}$
  - $\boldsymbol{r_E = L \times r_A - (L-1) \times r_D = r_D + L \times (r_A - r_D)}$
- **Effect of increasing leverage is** $\frac{\partial r_E}{\partial L} = r_A - r_D$
- Example 1: Leverage and Required Returns
  - A company sets its hurdle rate of ROE at least 15%. The following table describes some financial data of the bank. What is the minimum leverage need to achieve its goal?

Items	Amount
Return of asset	12%
Cost of debt	8%

  - Answer
    * $r_E = r_D + L \times (r_A - r_D)$
    * $15\% \leq 8\% + L \times (12\% - 8\%)$
    * $L \geq 1.75$
- Example 2: Leveraged Returns to Housing
  - Suppose house value is expected to rise 10% a year, neglecting the rental income and net of property maintenance costs. If the down payment of the house buyer is 20%, what is the annual rate of return to the homeowner by 6% loan rate?
  - Answer

$$\frac{\text{house price appreciation} - (1 - \text{down payment}) \times \text{loan rate}}{\text{down payment}} = \frac{10\% - (1 - 20\%) \times 6\%}{20\%} = 26\%$$

- Leverage in loans, options and derivatives
  - An economic balance sheet is construct to capture the implicit or embedded leverage in loans, options and derivatives.

3 Leverage: margin loans

- Margin lending has a straightforward impact on leverage.
- he haircut determines the amount of the loan that is made:
  - Haircut of h percent: (1- h) is lent against a given market value of margin collateral.
  - Leverage of h percent: The leverage on a position with a haircut of h percent is 1/h .

4 Leverage: short positions

- Lengthen the balance sheet: since both the value of the borrowed short securities and the cash generated by their sale appear on the balance sheet.
  - Increase leverage: both asset and liability increases the same amount with a constant equity.

$$L = \frac{A}{E} = 1 + \frac{D}{E}$$

5 Leverage: derivatives

- Derivative securities are a means to gain an economic exposure to some asset or risk factor without buying or selling it outright.
  - Mimic cash position: If the derivative creates a synthetic long (short) position, the economic balance sheet entries mimic those of a cash long (short) position.

6 Leverage: structured credit

- Seniority ranking: Bond tranche > Mezzanine tranche> Equity tranche.
  - The equity tranches get the remaining of the asset pool.
- Practice
  - On opening day, Master Fund LP has the following economic balance sheet: Assume Master Fund LP finances a long

Cash	$100 million
Financial assets	$200 million
Debt	$180 million
Equity	$120 million

position in $100 million worth of a stock at the margin requirement of 30%. It invests $30million of its own funds and borrows $70 million from a commercial bank. What is the change of firm's leverage ratio after trade?
  - Answer
    * 初始杠杆 $= \frac{asset}{equity} = \frac{300}{120} = 2.5$。
    * 交易后:
    * 资产 $= 70\,(cash) + 300\,(financial\ assets) = \$370$
    * 负债 $= 180\,(debt) + 70\,(margin\ loan) = \$250$
    * 权益 = 资产 - 负债 $= 370 - 250 = 120$
    * 交易后杠杆 $= \frac{asset}{equity} = \frac{370}{120} = 3.08$
    * 因此，杠杆从 2.5 变成 3.08。

2 Asset Liability Management

考点 1 The Investment and dealer Function In Financial-services Management Ch4&8

1 Investments: the crossroads account

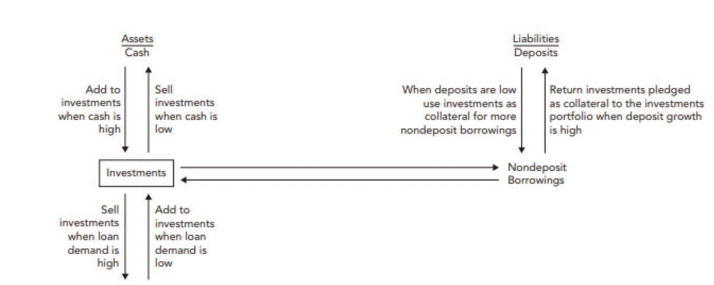


Exhibit 4.1 Investments: The crossroads account on a depository institution's balance sheet.

fig 2: instrument

2 Investment instruments

- To examine the different investment vehicles available, divide into two broad groups:
  - Money market instruments: maturity within one year , low risk and ready marketability.
  - Capital market instruments: maturities beyond one year, higher expected rate of return and capital gains potential.

3 Money market investment instruments

Instruments	Key Advantages	Key Disadvantages
Treasury bills	Safety and high liquidity; Good collateral for borrowing; Pledge behind government deposits;	Low yields relative to others; Taxable income;
Short-term Treasury notes and bonds	Safety; Good resale market; Good collateral for borrowing; Higher yield than T-bills;	More price risk than T-bills; Taxable gains and income;
Federal Agency Securities	Safety; Good to average resale market; Good collateral for borrowing; Higher yields than U.S. government securities;	Less marketable than treasury securities; Taxable gains and income;
Certificate Of Deposits (CDs)	Insured to at least \$100,000; Yields higher than on T-bills; Large denominations often marketable through dealers;	Limited resale market on longer-term CDs; Taxable income;
International Eurocurrency Deposits	Low risk; Higher yields than on many domestic CDs;	Volatile interest rates; Taxable income;
Bankers’Acceptances	Low risk due to multiple credit guarantees;	Limited availability at specific maturities; Issued in odd denominations; Taxable income;
Commercial Paper	Low risk due to high quality of borrowers;	Volatile market; Poor resale market; Taxable income;
Short-term Municipal Obligation	Tax-exempt interest income;	Limited resale market; Taxable capital gains;
Treasury Notes and Bonds	Safety and Good resale market; Good collateral for borrowing; May be pledged behind government deposits;	Low yields relative to long-term private securities; Taxable gains and income; Limited supply of longest-term issues;
Municipal Notes and Bonds	Tax-exempt interest income; High credit quality; Liquidity and marketability of selected securities;	Volatile market; Some issues have limited resale potential; Taxable capital gains;
Corporate Notes and Bonds	Higher pretax yields than on government securities; Aid in locking in higher long-term rates of return;	Limited resale market; Inflexible terms; Taxable gains and income;
Asset-backed securities (ABS)	Higher pretax yields than on Treasury securities; Collateral for borrowing additional funds;	Less marketable and more unstable in price than Treasury securities; May carry substantial default risk; Taxable gains and income;

table 1: Money and Capital Market Investment Instruments

4 Factors affecting choice of investment securities

- Expected Rate of Return:
  - YTM (if held to maturity) & holding period yield (HPY)

- Tax Exposure:
  - After-tax rate of return on loans and securities
  - Tax swap tool: lending institution sells lower-yielding securities at a loss to reduce taxable income and purchase new higher-yielding securities to boost future returns.
    - ∗ In years when loan revenues are high, it may be beneficial to engage in tax swapping.
  - Portfolio Shifting Tool (考虑 Tax and higher return):
    - ∗ 第一种操作: sell off selected securities at a loss
      Result: offset loan income and reduce tax liability
    - ∗ 第二种操作: shift portfolios to substitute new, higheryielding securities for old security whose yields may be below current market levels.
      Result: take substantial short-run losses in return for the prospect of higher long-run profits.
- Pledging Requirements
  - Depository institutions in the United States cannot accept deposits from federal, state, and local governments unless they post collateral acceptable to these governmental units to safeguard public funds.
    - ∗ Usually federal and municipal securities can meet government pledging requirements
- Risk
  - Credit or Default Risk
  - Liquidity Risk
  - Call Risk
  - Inflation Risk
  - Prepayment Risk
  - Business risk
  - Interest Rate Risk: Periods of rising (declining) interest rates are often marked by surging (decreasing) loan demand.

5 Investment maturity strategies

- Once the investments officer chooses the type of securities, there remains the question of how to distribute those security holdings over time.
- The ladder(or Spaced-Maturity)
  - Strategy: divide investment portfolio equally among all maturities.

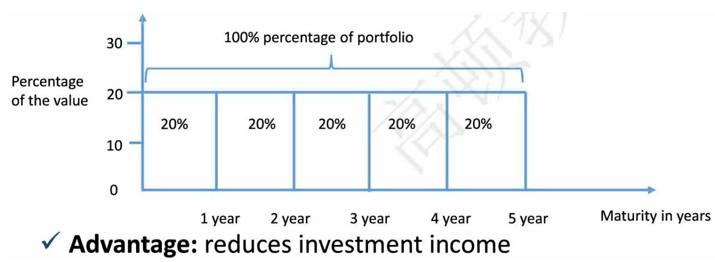


fig 3: ladder

  - Advantage: reduces investment income fluctuation/require little management expertise.
- The front-end load maturity policy

- Strategy: all security investments are short-term.

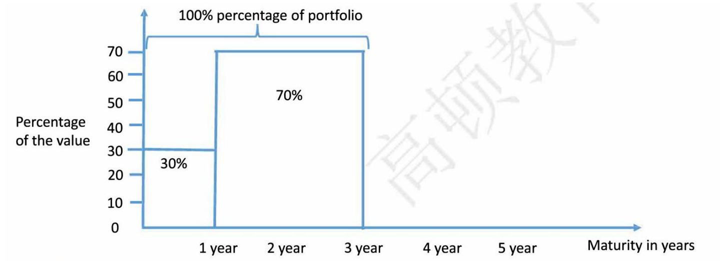


fig 4: front

- Advantage: strengthens the financial firm's liquidity position and avoids large capital losses if market interest rate rise.

• The back-end load maturity policy
- Strategy: all security investments are long-term.

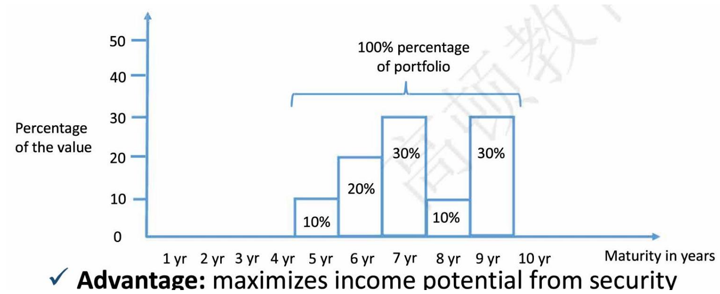


fig 5: back

- Advantage: maximizes income potential from security investments if market interest rate fall.

• The barbell investment portfolio strategy
- Strategy: security holdings are divided between short-term and long-term.

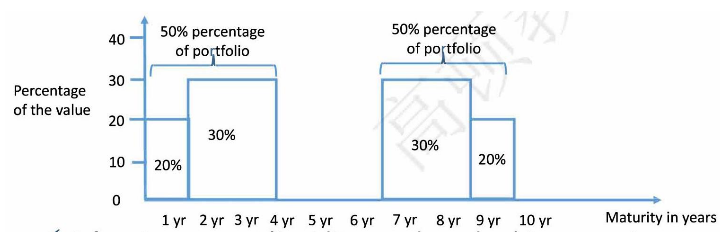


fig 6: barbell

- Advantage: meet liquidity needs and achieve earnings goals.

• The rate-expectations approach
- Strategy: change the mix of investment maturities as the interest rate outlook changes.

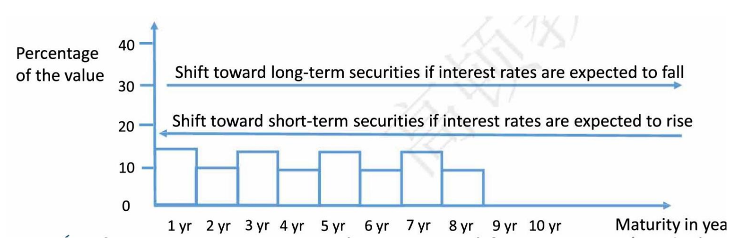


fig 7: rate

  - Advantage: maximizes the potential for earnings (and also for losses).

6 Maturity management tools

- The yield curve: a picture of how market interest rates differ across loans and securities of varying term (time) to maturity.
  - Forecast interest rates and the economy
  - Show current trade-offs between returns and risks
  - Measure how much might be earned currently by pursuing the carry trade
  - Riding the yield curve strategy: when the yield curve's slope is steep enough, buy the longer-term security (higher YTM) and hold until its tenor decreases (lower YTM), then sell it.
- Duration: present-value-weighted measure of maturity of an individual security or portfolio of securities.
  - Immunization of interest rate risk and reinvestment risk offset each other:
    * Duration of an individual security or a security portfolio is set equal to the investing institution's planned holding period.

7 Major lines of business of dealer banks

- Intermediate in Securities dealing, underwriting, and trading
  - Dealer 的基础业务: Intermediate in the primary market between issuers and investors of securities and in the secondary market among investors
  - Risk factors:
    * short-term funding like repo
- Trade in over-the-counter derivatives
  - For most over-the-counter derivatives trades, one of the two counterparties is a dealer.
    * Running matched book: aim to lay off much or all the risk of its positions by offset trades and made profits from the differences between bid and offer terms.
  - Proprietary trading in over-the-counter derivatives markets
  - Risk factors:
    * Contingent payments risk & Counterparty risk
- As Prime brokerage to hedge funds and other large investors
  - 主经纪商业务: Provides services including management of securities holdings, clearing, cash-management services, securities lending, financing, and reporting
  - 盈利方式: Generates additional revenues by lending securities that are placed by prime-brokerage clients
- Dealer banks have large asset-management divisions
  - Cater to the investment needs of institutional and wealthy individual clients.
- Prime brokerage and asset management
  - Risk factors: The dealer bank's liquidity position is weakened if large clients exit their positions or enter new positions to offset their risk
    * Systemic liquidity crisis

8 Failure mechanisms

- ”Run-on-the-bank”: The key mechanisms that lead to the failure of a dealer bank:
  - The flight of short-term creditors
  - The departures of prime-brokerage clients
  - Cash-draining actions by derivatives counterparties
  - Loss of clearing-bank privileges
- The flight of short-term creditors
  - Dealer bank 融资来源:
    - Bonds and commercial paper
    - Short-term repos
  - During financial crisis, repo rollover becomes difficult.
    - The dealer may be forced to sell its assets in a hurry, resulting in much lower prices. Lower price leads to further fire sales for others, causing a systemic risk.
- The way to mitigate the risk of flight of short-term creditors:
  - Establishing lines of bank credit
  - Dedicating a buffer stock of cash and liquid securities for emergency liquidity needs
  - ”Laddering” the maturities of its liabilities so that only a small fraction of its debt must be refinanced within a short period of time.
- The departures of prime-brokerage clients
  - Clients’ balances: The prime broker can use clients’ cash balances to meet the immediate cash demands of another.
  - Rehypothecation: Margin loans for a dealer bank can also be financed using the client’s own assets as collateral, through ”rehypothecation.”
  - Clients’ departures: If clients do leave without notice, their cash and securities are not in the pool, forcing dealer banks to use its own cash to meet liquidity needs.
- Cash-draining actions by derivatives counterparties
  - If a dealer bank is in a solvency crisis, OTC counterparty would look for opportunities to reduce exposure to that dealer bank.
    - Borrow from the dealer
    - Enter new trades that cause dealer to pay out cash
    - Seek to harvest cash from any derivatives positions
  - The way to mitigate the risk:
    - Central clearing counterparty
- Loss of cash settlement privileges
  - Daylight overdraft privileges: Under normal conditions, clearing bank extend ”daylight overdraft privileges” to creditworthy clearing customers.
  - Refuse the privileges: The clearing bank may refuse to process transactions that are insufficiently funded by the dealer bank’s cash fund account when the solvency of a dealer bank is questioned.

9 Policy responses to alleviate dealer's risk

- Capital injections: central banks and by capital injections into dealer banks.
  - Most important source of sy stemic risk is the potential impact of dealer-bank fire sales on market prices and investor portfolios.
- Higher capital requirements: capital are likely to be higher for derivatives that are not guaranteed by a central clearing counterparty.
- Central-bank insurance for tri-party repos: Short-term triparty repos are a particularly unstable source of financing in the face of concerns over a dealer's solvency.
- Central clearing: Threat by flight of OTC derivatives counterparties can be lowered by central clearing.
- Pre-failure resolution: such as distress-contingent convertible debt of dealer banks are important for those are suffering grievous financial distress.

考点 2 Managing Deposit Services and Non-deposit Liabilities Ch12& 13

1 Deposit Services

- Deposit accounts are the number one source of funds at most banks. Two key issues:
  - Where can funds be raised at lowest possible cost?
  - How can management ensure that the institution always has enough deposits to support lending and other services the public demands?

2 Deposit types

- Main type of deposit services types:
  - Transaction (payment or demand) deposits: honor any withdrawals made by the customer immediately.
  - Non-transaction (savings or thrift) deposits: Design to attract funds from customers who wish to set aside money for future expenditures or financial emergencies.

3 Transaction deposits

- Noninterest-bearing transaction (demand) deposits
  - Do not earn explicit interest but provide the customer with payment service.
  - Most volatile and least predictable, and with shortest potential maturity.
- Interest-bearing transaction deposits
  - Provide all the foregoing services and pay at least some interest.
  - Negotiable order of withdrawal (NOW) accounts
    - ∗ Hybrid checking-savings deposits
    - ∗ Give the offering depository institution prior notice before the customer withdraws funds
    - ∗ Held only by individuals and nonprofit institutions
  - Money market deposit accounts (MMDAs)
    - ∗ Flexible money market interest rates but accessible via check or preauthorized draft to pay for goods and services.
      - · Up to six preauthorized drafts and three withdrawals be made by writing checks per month.
    - ∗ Can be held by businesses as well as individuals.
  - Super NOWs

        * No limitation on the number of checks depositor write.
        * Offers institutions post lower yields on SNOWs than on MMDAs.
        * Can be held by individuals and nonprofit institutions.
    - Mobile apps
        * The hottest item in the transaction deposit field today appears to be the mobile check deposit.

•

4 Non-transaction deposits

- Passbook saving deposits
    - Unlimited withdrawal privileges and low interest rate
    - Tend to be stable with little sensitivity to changes in interest rates.
    - Individuals, nonprofit organizations, governments, and business firms can hold savings deposits
        * Businesses could not place more than $150,000 in the United States
- Time deposits
    - Provide to wealthier depositors with fixed maturity dates and fixed or sometimes fluctuating rates.
    - Must carry a minimum maturity of 7 days.
    - Most popular of all time deposits is certificates of deposits (CDs)
        * Negotiable CD and nonnegotiable CD.
- Retirement savings deposits
    - Wage earners and salaried individuals: individual retirement account （IRA）grant the right to make limited tax-free contributions each year.
    - Self-employed persons: Keogh plan retirement account is available.
    - Both IRAs and Keoghs are highly stable and carries fixed interest rate, which are great appeal for managers of depository institutions.

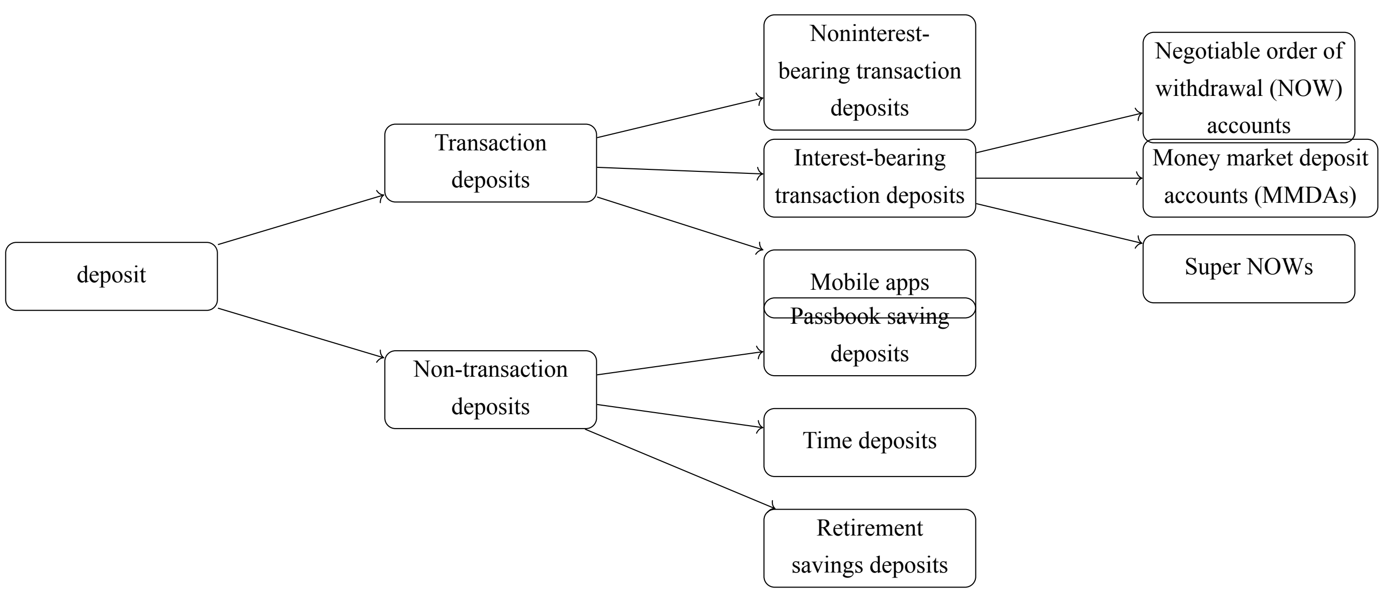


fig 8: 存款类型结构图

5 Pricing Deposits

- An important issue remains: How should depository institutions price their deposit and deposit services in order to attract new funds and make a profit?
  - Need to pay a high enough interest return to attract customer funds but must avoid paying an interest rate so costly that it erodes any potential profit margin.

6 Deposits and Deposit Services Pricing Methods

- Marginal Cost
- Cost plus profit margin
- Conditional Pricing
- Relationship pricing

7 Deposits Pricing Methods

- Marginal Cost
  - Definition: the added cost of bringing in new funds.
  - Formular: Marginal cost rate = $\frac{Changeintotalcost}{Addictionalfundsraised}$
    * Change in total cost (Marginal cost)= New interest rate × Total funds raised at new rate - Old interest rate × Total funds raised at old rate
  - Profit maximum
    * marginal revenue rate = marginal cost rate
- Example
  - New Day Bank plans to launch a new deposit campaign next week in hopes of bringing in from $100 million to $600 million new deposit, which it expects to invest at a 4.05 percent yield. The following chart is the cost for attracting new deposits.

New Deposits (in millions)	100	200	300	400	500	600
Interest cost	2%	2.25%	2.5%	2.75%	3%	3.25%

  - What interest rate should the institution set to ensure that profit is maximized?
  - Answer:

Expected Amounts of New Deposits That Will Flow In	Average Interest the Bank Will Pay on New Funds	Total Interest Cost of New Funds Raised	Marginal Cost of New Deposit Money	Marginal Costs as a Percentage of New Funds	Total Profits Earned (after interest cost)
100	2%	2	2	2%	2.05
200	2.25%	4.5	2.5	2.5%	3.60
300	2.5%	7.5	3	3%	4.65
400	2.75%	11	3.5	3.5%	5.20
500	3%	15	4	4%	5.25
600	3.25%	19.5	4.5	4.5%	4.80

  - The institution should set deposit interest rate at 3%

8 Deposit Services Pricing Methods

- Cost plus profit margin
  - Each deposit service may be priced high enough to recover all or most of the cost of providing that service.
  - Calculate formula:

$$\begin{aligned}\text{Unit price charged the customer for each deposit service} &= \text{Operating expense per unit of deposit service}\\ &+ \text{Estimated overhead expense allocated to deposit service}\\ &+ \text{Planned profit margin from each service unit sold}\end{aligned}$$

- Example
  - G&F Savings Bank finds that its basic transaction account, which requires a $1000 minimum balance, costs an average of $3.25 per month in servicing costs (including labor and computer expense) and $1.25 per month in overhead expenses. The bank also tries to gain a $0.50 per month profit margin on these accounts. What monthly fee should the bank charge each customer?
  - Answer:
    - $3.25 + $1.25 + $0.50 = $ 5
- Conditional Pricing
  - The customer pays a low fee or no fee if the deposit balance remains above some minimum level but faces a higher fee otherwise.
  - Varies among deposits based on one or more factors as follows:
    - The number of transactions
    - The average balance
    - The maturity of the deposit
  - Flat-rate pricing
    - The depositor's cost is a fixed charge per check, per time period, or both.
  - Free pricing
    - No monthly account maintenance fee or per-transaction charge.
    - Unprofitable for institutions because it tends to attract many small, active deposits when market rate is high.
  - Conditionally free pricing
    - Favors large-denomination deposits because services are free if the account balance stays above some minimum figure.
- Relationship pricing
  - Promotes greater customer loyalty
  - makes the customer less sensitive to the prices posted on services offered by competing financial firms.

Bank A		**Bank B**	
Regular checking account:		*Regular checking account:*	
Minimum opening balance	$100	Minimum opening balance	$100
If minimum daily balance is		If minimum daily balance is	
$600 or more	No fee	$500 or more	No fee
$300 to $599	$5.00 per mo.	Less than $500	$3.50 per mo.
Less than $300	$10.00 per mo.		
If the depositor's collected monthly balance averages $1,500, there is no fee		If checks written or ATM transactions (debits) exceed 10 per month and balance is below $500	$0.15 per debit
No limit on number of checks written			
Regular savings account:		*Regular savings account:*	
Minimum opening balance	$100	Minimum opening balance	$100
Service fees:		Service fees:	
If balance falls below $200	$3.00 per mo.	If balance falls below $100	$2.00 per mo.
Balance of $200 or more	No fee	Balance above $100	No fee
Fee for more than two		Fee for more than three	
withdrawals per month	$2.00	withdrawals per month	$2.00

Exhibit 12.1 **Example of the use of conditional deposit pricing by two banks serving the same market area.**

fig 9: DEPOSITPRICE

- Deposit insurance
  - 供保机构: Federal Deposit Insurance Corporation (FDIC) provides deposit protection for its membership.
  - 免责范围: U.S. government securities, shares in mutual funds, safe deposit boxes, and funds stolen from an insured depository institution.
  - 保费制定: determined by the volume of deposits and the degree of risk exposure.
- Disclosure of deposit service
  - Advertising of deposit terms may not be misleading.
  - Deposit rates must be quoted as annual percentage yields(APY).
  - The dates and minimum balance required must be explicit.
  - The depositor must be warned of penalties or fees that could reduce the yield.

9 Overdraft protection

- Definition: This service makes sure that clients' incoming checks and draft are paid and clients avoid excessive NSF (not sufficient funds) fees.
  - Provide customers convenience but like predatory lending
  - High actual interest rate: Important source of income for financial-service provider.
- Being criticized: Faced with adverse public and regulators' comments, some financial firms have reduced or eliminated their overdraft fees.

10 Basic (Lifeline) Banking

- Controversial social issue: Depository institutions are asked to offer low-cost financial services, especially deposits and loans, for customers unable to afford conventional services.
  - Controversial for service provider:
    * Low price cannot cover cost: Most financial-service providers are privately owned corporations and responsible to their shareholders to earn profit.
    * Regulation problem: banking industry is regulated and are not treated in public policy like other private firms.
  - Controversial for solving the issue:
    * How to decide which customers access to low-price services?
    * Insist on imposing a means test on customer?
    * Someone must bear the cost of producing services but who?

11 Practices

- First Metrocentric Bank posts the following schedule of fees for its household and small-business transaction accounts: For average monthly account balances over $1,500, there is no monthly maintenance fee and no charge per check or other draft. For average monthly account balances of $1,000 to $1,500, a $2 monthly maintenance fee is assessed and there is a $1 charge per check or charge cleared. For average monthly account balances of less than $1,000, a $4 monthly maintenance fee is assessed and there is a $1.5 per check or per charge fee. What form of deposit pricing is this?
  A. Cost plus profit margin
  B. Marginal Cost
  C. Conditional Pricing
  D. Relationship pricing
- Answer: C
  解析: 根据题目描述，可以得知不同的存款余额的条件下，对于账户的收费是不一样的，属于条件定价。

12 Sources of Non-deposit Liabilities

- When deposit is inadequate to support loans and investment, financial institutions need non-deposit liabilities:
  - Federal Funds Market (Most popular domestic source)
  - Borrowing from Federal Reserve Banks
  - Advances from Federal Home Loan Banks
  - The large Negotiable CDs
  - The Eurocurrency Deposit Market
  - Commercial Paper market
  - Repurchase Agreements
  - Long-term non-deposit funds
- Federal Funds Market (”Fed Funds”): Consist exclusively of deposits held at the Federal Reserve banks and can be transferred through Fedwire.
  - Meet legal reserve requirements and satisfy loan demand
  - Large banks play a role similar to funds brokers.
  - Short-term borrowings:
    * Overnight loans: Normally no collateral

    * Term loans: longer-term Fed loans contract
    * Continuing contract: automatically renewed each day

- Borrowing from Federal Reserve Banks: Fed will make the loan through its discount window by crediting the borrower's reserve account. The loan must be backed by acceptable collaterals.
  - Primary credit: short terms(overnight to 90 days), a higher interest rate than Fed funds interest rate.
  - Secondary credit: depository institutions not qualifying for primary credit, a higher interest rate than primary credit.
  - Seasonal credit: longer periods than primary credit, for small and medium-sized institutions experiencing seasonal swings.
- Advances from Federal Home Loan Banks (FHLB)
  - 目标: Improves the liquidity of home mortgages and encourages more lenders to provide credit to the housing market.
  - 提供纾困抵押贷款: Allows troubled institutions to use the home mortgages as collateral for emergency loans.
  - 贷款利率低于市场利率: A stable source of funding at below-market interest rates.
  - 期限范围广: Ranges from overnight to more than 20 years
- The large Negotiable CDs
  - A hybrid account: a deposit or another form of IOU.
  - Short maturity: range from 7 days to 1 or 2 years but concentrated in the 1- to 6-month.
  - 到期本息和的计算： Amount due CDs = Principal + Principal $\times \frac{Days\ to\ maturity}{360\ days} \times$ Annual interest rate
- The Eurocurrency Deposit Market
  - How to borrow and lend Eurocurrencies: A domestic financial firm can tap the Euromarkets for funds by contacting one of the major international banks that borrow and lend Eurocurrencies every day.
    * The largest U.S. bank use their own overseas branches tap the market.
  - The largest unregulated financial market-place in the world.
- Commercial Paper market
  - Short-term notes: 3 or 4 days to 9 months, sell at discount.
  - Industrial paper: designed to finance the purchase of inventories or raw materials and to meet other immediate cash needs of nonfinancial companies.
  - Financial paper: issued by finance companies and the affiliates of financial holding companies.
    * Purchase off-book loans of other financial firms in same organization, giving these firms funds to make new loans.
- Repurchase Agreements
  - Less popular than Fed funds and more complex.
  - Most domestic Repos are transacted across the Fed Wire system, just as are Fed funds transactions.
  - A bit longer time to transact than a Fed funds loan.
  - Repo transaction is often for overnight funds, but it may be extended for days, weeks, or even months.
- Long-term non-deposit funds
  - Beyond one year: 5 to 12 years mortgages, capital notes and debentures.
  - Favorable leveraging effect: made it attractive to larger financial firms.
  - More sensitive: Tend to be a sensitive barometer of the perceived risk exposure.
-

13 Non-deposit Funding Choice

- In using non-deposit funds, funds managers must answer the following key questions:
  - 1. How much in total must be borrowed from these sources to meet funding needs?
    - Available Funds Gap
  - 2. Which non-deposit sources are best, given the borrowing institution's goals?
    - The relative costs, risk, length of time, size of the institution, regulations limits.

14 Available Funds Gap (AFG)

- Calculation formular:
  -

$$\begin{aligned}\text{Available funds gap (AF)} = &\ \text{Current and projected loans and investments the} \\ &\ \text{lending institution desires to make} \\ &- \text{Current and expected deposit inflows and other} \\ &\ \text{available funds}\end{aligned}$$

  - Small amount add to AFG: cover unexpected credit demands or unanticipated shortfalls in inflow funds.
- Example
  - Suppose a commercial bank has new loan request that meet its quality standards for $150 million, it wishes to purchase $75 million in new Treasury securities and expects drawings on credit lines from its customers of $135 million. Deposit and other customer funds received today total $185 million, and those expected in the coming week will be $100 million.
  - Answer:
    - ∗ AFG= ($150 + $75 + $135) - ($185 + $100) = $75m

15 Factors of choosing alternative non-deposit sources

- The relative costs of raising funds from each source.
- The risk (volatility and dependability) of each funding source.
- The length of time (maturity or term) for which funds are needed.
- The size of the institution that requires funds.
- Regulations limiting the use of alternative funds sources.
- The relative costs of raising funds
  - Effective cost rate: determine how much each source of borrowed funds costs.
- The relative costs of raising funds from each source

$$\text{Effective cost rate on funds} = \frac{\text{(Current interest cost on amounts borrowed+ Noninterest costs incurred to access these funds)}}{\text{Net investable funds raised from this source}}$$

$$\text{Current interest cost on amounts borrowed} = \text{Prevailing interest rate in the money market} \quad \times \text{Amount of funds borrowed}$$

  - Noninterest costs to access funds = (Estimated cost rate representing staff time, facilities, and transaction costs) x Amount of funds borrowed

  - Net investable funds = Total amount borrowed less legal reserve requirements (if any), deposit insurance (if any), and funds placed in nonearning assets
  - Deposit insurance = Total deposits received from the public x Insurance fee per dollar

- Example: Effective cost rate
  - Suppose that Fed funds are currently trading at an interest rate of 6.0 percent. Moreover, management estimates that the marginal noninterest cost, in the form of personnel expenses and transactions fees, from raising additional monies in the Fed funds market is 0.25 percent. Suppose that a depository institution will need $25 million to fund the loans it plans to make today, of which only $24 million can be fully invested due to other immediate cash demands. What is the effective cost rate Fed funds ?
  - Answer
    Current interest cost on Federal funds = 0.06 x $25m = $1.5m
    Noninterest cost to access Federal funds raised = 0.0025 x $25m = $0.063m

$$\begin{aligned}\text{The effective annualized Fed funds cost rate} &= \frac{\$1.5\text{ m} + \$0.063\text{ m}}{\$24\text{ m}} \\ &= 6.51\%\end{aligned}$$

- Example: Effective cost rate considering insurance fee
  - Suppose management decides to consider borrowing funds by issuing negotiable CDs that carry a current interest rate of 7.00 percent. Moreover, raising CD money costs 0.75 percent in noninterest costs. An additional expense associated with selling CDs to raise money is the deposit insurance fee, assume the current insurance rate is $0.0027 per dollar of deposits. What is the effective cost rate of negotiable CDs?
  - Answer

$$\begin{aligned}\text{Current interest cost on negotiable CDs} &= 0.07 \times \$25\text{m} = \$1.75\text{m} \\ \text{Noninterest cost to negotiable CDs} &= 0.0075 \times \$25\text{m} = \$0.1875\text{m} \\ \text{Insurance fee} &= \$25\text{ million} \times 0.0027 = 0.0675\text{m} \\ \text{Effective cost rate} &= \frac{(1.75\text{m} + 0.1875\text{m})}{\$24\text{ million} - \$0.0675\text{ million}} = 8.10\%\end{aligned}$$

- The risk (Volatility and dependability) of each funding source.
  - Interest rate risk: The volatility of credit costs, the shorter the term of the loan, the more volatile the prevailing market interest rate tends to be.
  - Credit availability risk: Negotiable CD, Eurodollar, and commercial paper markets are especially sensitive to credit availability risks.
- The length of time (maturity and term) funds are needed
  - Some funds sources, such as commercial paper and longterm debt capital may be difficult to access immediately.
  - Term or maturity of the funds need plays a key role as well.
- The size of the borrowing institution
  - The standard trading unit for most money market loans is $1 million.
  - The Fed funds and central bank's discount window can make small denomination loans for smaller firms.
- Regulations limiting the use of alternative funds sources
  - Federal and state regulations may limit the amount, frequency, and use of borrowed funds.

* In the United States CDs must be issued at least 7 days.
* The Federal Reserve banks may limit borrowings from the discount window, particularly by depository institutions that appear to display significant risk of failure.
* Other forms of borrowing may be subjected to legal reserve requirements by action of the central bank.

16 Overall cost of funds calculation

- The relative costs of raising funds from various sources including deposits
  - Historical average cost approach (look at the past)
    * Weighted average interest expense
    * Break-even cost (cover the operating and interest cost)
    * Weighted average overall cost of capital
  - Pooled-funds approach (look at the future)
    * Pooled deposit and nondeposit funds expense rate
    * Hurdle rate
  - Example: Historical Average Cost Approach
    * A bank needs to raise fund for future loans, it asks what funds the financial firm has raised to date and what they cost:

Sources of Funds Drawn	Average Amount of Funds Raised (millions)	Average Rate of Interest Incurred
Noninterest - bearing demand deposits	\$100	0%
Interest - bearing transaction deposits	\$200	7%
Savings accounts	\$100	5%
Time deposits	\$500	8%
Money market borrowing	\$100	6%

    * Assumed the operating costs is \$10 million and all earning assets is \$750 million, what is the weighted average interest expense and break-even cost rate on borrowed funds invested in earning assets? Assumed the required rate of return of \$100 million investment from stockholders is 12% after tax (tax rate is 35%), what is the weighted average overall cost of capital?
    * Answer

Sources of Funds Drawn Upon	Avg. Amount (millions)	Avg. Interest Rate	Interest Paid (millions)
Noninterest-bearing demand deposits	\$100	0%	\$0
Interest-bearing transaction deposits	\$200	7%	\$14
Savings accounts	\$100	5%	\$5
Time deposits	\$500	8%	\$40
Money market borrowing	\$100	6%	\$6
Total funds raised = \$1,000		All interest costs = \$65	

table 2: Summary of Fund Sources and Associated Interest Costs

$$\text{Weighted average interest expense} = \frac{\text{All interest paid}}{\text{Total funds raised}} = \frac{65}{1,000} = 6.5\%$$

$$\text{break - even cost rate} = \frac{\text{Interest} + \text{Other operating costs}}{\text{All earnings assets}} = \frac{65+10}{750} = 10\%$$

Weighted average overall cost of capital

$$= \text{Break - even cost on borrowed funds} + \frac{\text{After - tax cost of stockholders}'\text{investment}}{1 - \text{Tax rate}} \times \frac{\text{Stockholders}'\text{investment}}{\text{Earning assets}}$$

$$= 10\% + \frac{12\%}{1-0.35} \times \frac{100}{750} = 12.5\%$$

* Tips: 12.5% is the lowest rate of return over all fund-raising costs and the borrowing institution can afford to earn on its assets if its equity shareholders invest $100 million in the institution.

- Example: Pooled-funds approach
  * A bank needs to raise for future loans. Suppose the estimate for future funding sources and costs is as follows:

Profitable New Deposits and Nondeposit Borrowings	Dollars of New Deposit & Borrowing (millions)	Portion Placed in New Earning Assets	Interest & Other Operating Expenses
Interest-bearing transaction deposits	$100	50%	8%
Time deposits	$100	60%	9%
New stockholders' investment	$100	90%	13%

table 3: Summary of New Deposit and Investment Sources

What is the Pooled deposit and non-deposit funds expense rate and what is the hurdle rate of return over all earning assets ?

* Answer

Profitable New Deposits and Nondeposit Borrowings	Dollars of New Deposit & Non-deposit Borrowing (millions)	Portion of New Borrowings Placed in New Earning Assets	Dollar Amount Placed in Earnings Assets (millions)	Interest Expense & Other Operating Expenses of Borrowing	All Operating Expense Incurred (millions)
Interest - bearing transaction deposits	$100	50%	$50 = 100 × 50%	8%	$8 = 100 × 8%
Time deposits	$100	60%	$60 = 100 × 60%	9%	$9 = 100 × 9%
New stockholders'investment in the institution	$100	90%	$90 = 100 × 90%	13%	$13 = 100 × 13%
Total	$300		$200		$30

.

$$\text{Pooled deposit and nondeposit funds expense rate} = \frac{\text{All expected operating expenses}}{\text{All new funds expected}} = \frac{\$30}{\$300} = 10\%$$

.

$$\text{Hurdle rate of return over all earning assets} = \frac{\text{All expected operating expenses}}{\text{Dollars available to place in earnings assets}} = \frac{\$30}{\$200} = 15\%$$

Tips: Hurdle rate of return over all earning assets: minimum rate of return must be earned on any future loans and investments just to cover the cost of all new funds raised.

考点 3　Repos And Financing Ch14

1　Repurchase Agreements Introduction

- A contract in which a security is traded at some initial price with the understanding that the trade will be reversed at some future date at some fixed price.
  - Repo rate: actual/360, simple rate
  - Repo haircut: difference between the purchase price of the securities and the amount borrowed from the repo.
  - Collateral: investors care about the quality of the collateral
  - Margin call: borrower supplies extra collateral in declining markets but may withdraw collateral in advancing markets.

2　Repurchase Agreements calculation

- Interest cost of Repo = Amount borrowed x Repo rate x Number of days in Repo borrowing /360 days
- Amount borrowed(cash inflow) = collateral securities full price (invoice price) x (1 - haircut)
- Cash outflow = Amount borrowed + Interest cost of Repo

3　Example

- Bank Venus buys a bond on its coupon payment date. Three months later, in order to generate immediate liquidity, the bank decides to repo the bond with Bank Lu. Details of the bond and repo transaction are in the following chart. If the repo contract expires 6 months (184 days) from now, what is the bank's expected cash outflow at the end of the repo transaction?

Notional	(USD) 1000,000
Coupon (semi - annual)	6%
Current quote bond price	(USD) 97.5
Repo haircut	5%
Repo interest rate	1.05%

- Answer
  - Cash inflow at beginning of repo:

$$1000000 \times (97.5\% + 6\% \times 0.25) \times (1 - 5\%) = 940500$$

  - Cash outflow at end of repo:

$$940500 \times \left(1 + 1.05\% \times \frac{184}{360}\right) = 945547.35$$

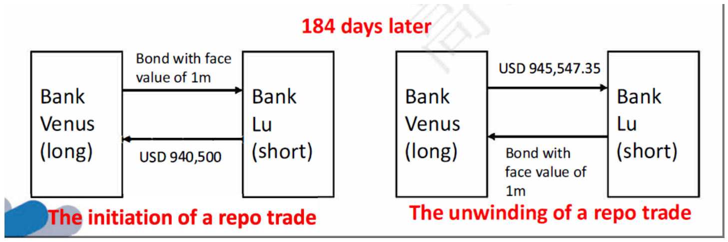


fig 10: cash

4 Common motivations for investing repo

- Reverse Repos and Short Positions
  - Buy the repo/ enter into reverse repo/ invest in repo: lender of cash in repo market
    - ∗ Professional investors often want to short a bond.
      - · Bet the interest rates will rise
      - · Bet that the price of another security will rise relative to the price of the security being sold short.
- Reverse Repos and Cash Management
  - Investors holding cash for liquidity or safekeeping purposes often find investing in repo to be an ideal solution.
    - ∗ Money market mutual fund industry: invests on behalf of investors willing to accept low returns in exchange for liquidity and safety.
    - ∗ Municipalities: These tax revenues cannot be invested in risky securities, but neither should the cash collected lie idle.
- Repos and Long Financing
  - Sell the repo/enter into repo: borrowers of cash in repo markets.
    - ∗ Financing for proprietary positions/customers positions.
    - ∗ Finance inventory for the purpose of making markets.
      - · open repo: a one-day repo that renews itself day-today until cancelled by either party.
      - · Back-to-back repo

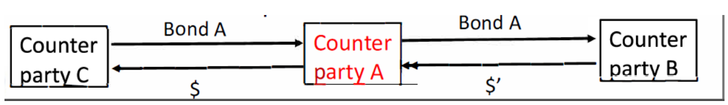


fig 11: tp

- Repos and Liquidity Management
  - Firms balance the costs of funding against the risks of being caught without the financing necessary to survive.
    - ∗ Repo markets are relatively liquid
    - ∗ Repo borrowing rates relatively low
    - ∗ Short maturities repo is on the less stable side of the funding spectrum, although more stable than short-term, unsecured borrowing like commercial paper.

5 Risk for entering repo

- Counterparty risk: borrower defaults and the collateral's value turns out to be insufficient to cover the loan amount
  - Overnight repo and high quality collateral is preferred
- Liquidity risk: the value of the collateral declining in fire sales
  - can be mitigated by haircut

6 Repo Financing and the Collapse of Bear Stearns

- Case brief: During 2006, Bear Stearns decided to reduce short-term unsecured funding and increased the average term of its secured funding during the first half of 2007. Bear Stearns was able to obtain longer term repo facilities of six months or more to finance assets such as whole loans and non-agency mortgage-backed securities. During 2007-2008 crisis, repo lenders began shortening the duration and require higher quality collateral. In the week of March 10, 2008, Bear Stearns suffered a run.
- Role of repo in the collapses of Bear Stearns:
  - The loss of confidence caused three related consequences:
    * Withdrew increase: Prime brokerage clients withdrew cash and unencumbered securities at a rapid rate.
    * Stop renewing repo: Repo lenders declined to roll over repo loans, even when the loans were supported by high-quality collateral such as agency securities.
    * Capital flight: Result in a rapid flight of capital from the firm that could not be survived.

7 Repo Financing and the Collapse of Lehman Brothers

- Case brief: JPMorgan Chase(JPM) was Lehman's tri-party repo clearing agent. Before Lehman's final week, JPM's tri-party lending to Lehman typically exceeded $100 billion. JPM had historically not taken any haircuts on its tri-party, intraday advances, but began to do so in early 2008. In Lehman's case, JPM phased in haircuts so as to match, by mid-August 2008, the haircuts posted to overnight repo investors. Finally, Lehman Brothers run out of liquidity to bankrupt.
- Role of repo in the collapses of Lehman Brothers:
  - Over rely on repo: Lehman Brothers relied heavily on shortterm repo as a financing tool.
  - Haircut increase: When its tri-party repo clearing agent JPM increased haircut, Lehman Brothers could not meet and finally run out of liquidity and finally got bankrupted.

8 General and special repo rates

- Repo trades can be divided into those using general collateral (GC) and those using special collateral or specials.
  - GC rates are for overnight repo where any U.S. Treasury collateral is acceptable.
  - Special rates are for repo being able to borrow cash at a relatively low rate with special securities as collateral.
  - Special rate is less than the GC rate:
    * Special spreads = C rates - Special rates

9 Special spreads and auction cycle

- Special collateral:
  - Special rates determined by the demand of borrowing that issue to that date relative to the supply available.
  - On-the-run (OTR) or current issue: The most recently issued bond of a given maturity
    * At each maturity, the OTR trades the most special, followed by the old, the double-old.
- Example: 10- and 30-year U.S. Treasury bonds along with overnight repo rates and spreads as of May 28, 2010
- Why OTR securities tend to trade special: liquid is special
  - OTR tend to be the most liquid: Extra liquidity of newly issued Treasuries makes them ideal candidates not only for long positions but for shorts.
    * Self-fulfilling phenomenon: everyone expects a recent issue to be liquid.
- Four characteristics of special spreads:

General collateral rate 0.23%					
coupon	maturity	Issue date	designation	Repo rate	spread
$3\frac{1}{2}$%	5/15/20	5/17/10	OTR 10yr	0.09%	0.14%
$3\frac{5}{8}$%	2/15/20	2/16/10	Old 10yr	0.21%	0.02%
$3\frac{3}{8}$%	11/15/19	11/16/09	Dbl - Old 10yr	0.21%	0.02%
$4\frac{1}{2}$%	5/15/40	5/17/10	OTR 30yr	0.18%	0.05%
$4\frac{5}{8}$%	2/15/40	2/16/10	Old 30yr	0.20%	0.03%
$4\frac{3}{8}$%	11/15/39	11/16/09	Dbl - Old 30yr	0.21%	0.02%

- Quite volatile: spreads are volatile on a daily basis
- Quite large: spreads of hundreds of basis points are quite common.
- Higher levels over some periods: Special spreads do attain higher levels over some periods rather than others.
- Small immediately after auctions and to peak before auctions: The supply of high liquid bond is high after auctions and low before auctions.

10 Financing Value of a Bond Trading Special in Repo

- Financing value of a bond: over the entire life of the bond, borrow cash at its special rate, and invest that cash at the higher GC rate.

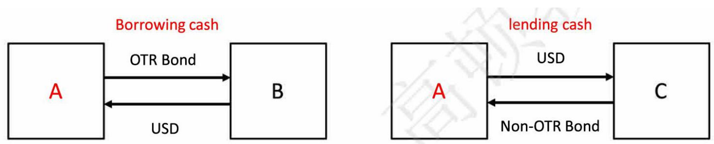


fig 12: OTR

  - Financing value of a bond tading special repo = Amount borrowed x special spread x Number of days in Repo borrowing /360 days

- Example
  - Assume that the GC rate is 0.25% and the an OTR bond is currently treated as special securities and will be treated as general collateral from September 30, 2022. The special rate of this bond is only one basis point. Assuming no haircut, What is the financing value per $100 market value of the bond over the 125 days from the pricing date, May 28, 2022, to September 30, 2022.
- Answer
  - Financing value of a bond tading special repo 125 = 100 X (0.25% - 0.01 %) X 360 = 0.083
  - It means that the financing value is 8.3 cents per 100 dollar market value of the bond.

考点 4 Risk Management For changing Interest Rates Ch18

1 Asset-liability Management Strategy

- A balanced approach, also called funds management strategy.
  - Manage both assets and liabilities in order to achieve the financial institution's goals.

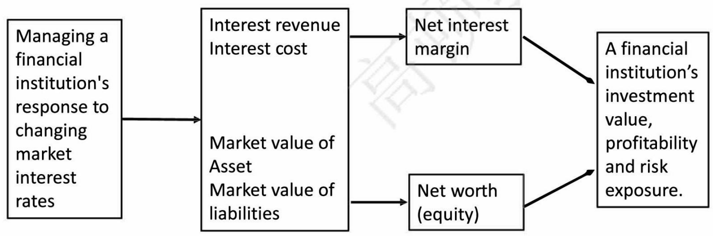


fig 13: goals

2 The impact of interest rate risk

- Interest rate risks faced by financial institutions affect both the balance sheet (Net worth) and the statement of income and expenses (Net interest margin).
- When interest rates change in the financial marketplace:

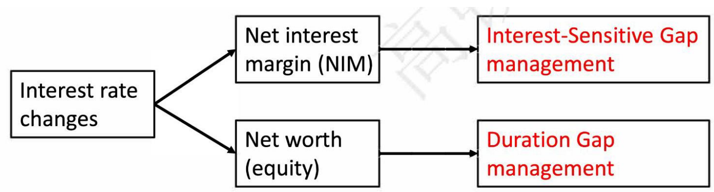


fig 14: irc

3 Net interest margin

- Formular: Net interest margin (NIM) = $\frac{\text{Net interest income}}{\text{Total earning assets}} = \frac{\text{Interest income}-\text{Interest expense}}{\text{Total earning assets}}$
  - Interest income from loans and investments
  - Interest expense from deposits and other borrowed funds
- Types of assets and liabilities
  - Repriceable (interest-sensitive) assets/liabilities
    * Variable-rate loans and securities etc.
  - Nonrepriceable assets/liabilities
    * Fixed interest rate loans etc.
- Net interest income = Total interest income - Total interest cost
  - Total interest income = $Y_{ISA} \times Q_{ISA} + Y_{Non\ ISA} \times Q_{Non\ ISA}$
  - Total interest cost = $Y_{ISL} \times Q_{ISL} + Y_{Non\ ISL} \times Q_{Non\ ISL}$
    * $Y_{ISA}(Y_{ISL})$: average interest yield (cost) on rate sensitive assets (liabilities)
    * $Q_{ISA}(Q_{ISL})$: volume of rate sensitive assets (liabilities)
    * $Y_{non\ ISA}(Y_{Non\ ISL})$: average interest yield (cost) on fixed (nonsensitive)assets(liabilities)
    * $Q_{non\ ISA}(Q_{Non\ ISL})$: volume of fixed assets(liabilities)
- Example
  - Suppose the yields on interest rate sensitive and fixed assets average 10% and 11%, respectively, while interest rate sensitive and non-interest rate sensitive liabilities cost an average of 8% and 9%, respectively. During the coming week the bank holds $1,700 million in interest rate sensitive assets (out of an asset total of $4,100 million) and $1,800 million in interest rate sensitive liabilities. Please calculate the net interest income assuming the equity is zero.
  - Answer
    * Net interest income = 0.1 X 1,700 + 0.11 X (4,100 - 1,700) - 0.08 X 1,800 - 0.09 x (4,100 - 1,800) = $83 million

4 Interest-Sensitive Gap management

- Interest-sensitive gap(IS GAP) = Interest-sensitive assets （ISA） - Interest-sensitive liabilities(ISL)
- Relative IS GAP = IS GAP/Size of financial institution
- Interest Sensitivity Ratio (ISR) = ISA/ISL
- Asset-sensitive (positive) gap: ISA - ISL > 0
  - If interest rates rise, bank's NIM will increase.
- Liability-sensitive (negative) gap: ISA - ISL < 0
  - If interest rates rise, bank's NIM will decrease.
- Zero interest-sensitive gap: means net interest margin is protected regardless of which way interest rates will go.
  - Does not eliminate all interest rate risk.
- Aggressive GAP management

Expected Changes in Interest Rates (Management's Forecast)	**Best Interest-Sensitive GAP Position to Be in:**	**Aggressive Management's Most Likely Action**
Rising market interest rates	Positive IS GAP	Increase interest-sensitive assets Decrease interest-sensitive liabilities
Falling market interest rates	Negative IS GAP	Decrease interest-sensitive assets Increase interest-sensitive liabilities

- Defensive Interest-Sensitive Gap management
  - Set interest-sensitive GAP as close to zero as possible to reduce the expected volatility of net interest income.

5 Limitation of Interest-sensitive Gap management

- Changes inconsistent: Interest rates paid on liabilities tend to move faster that interest rates earned on assets.
- Basis risk: The interest rates attached to assets of various kinds often change by different amounts and at different speeds than many of the interest rates attached to liabilities.
- Net worth is not considered.

6 Duration Gap management

- Duration analysis can be used to stabilize , or immunize, the market value of a financial institution's net worth (NW).
- NW = Assets - Liabilities
  - A rise in market rates of interest will cause the market value (price) of both fixed-rate assets and liabilities to decline.
- Price sensitivity to interest rates changes and duration
  - $\frac{\Delta P}{P} \approx -D * \frac{\Delta i}{(1+i)}$, D : Macauley duration
- $\Delta NW = \Delta A - \Delta L = \left[-D_A \times \frac{\Delta i}{(1+i)} \times A\right] - \left[-D_L \times \frac{\Delta i}{(1+i)} \times L\right] = -\left[D_A - D_L \times \frac{L}{A}\right] \times \frac{\Delta i}{(1+i)} \times A$
  - $D_A - D_L$: Duration gap
  - $D_A - D_L \times \frac{L}{A}$: Leverage - adjusted duration gap
  - D_A: Dollar - weighted duration of asset portfolio
  - D_L: Dollar - weighted duration of liability portfolio

Leverage - adjusted duration gap	Market interest	The Financial Institution's Net Worth Will:	Value change of Asset/Liability
Positive	Rise	Decrease	Assets value ↓↓ Liabilities value ↓
	Fall	Increase	Assets value ↑↑ Liabilities value ↑
Negative	Rise	Increase	Assets value ↓ Liabilities value ↓↓
	Fall	Decrease	Assets value ↑ Liabilities value ↑↑
Zero	Rise	No change	Assets value ↓ Liabilities value ↓
	Fall	No change	Assets value ↑ Liabilities value ↑

- Portfolio immunization strategy
  - Leverage-adjusted duration gap equals to zero
- Take chance to maximize the shareholders' position

Expected Change in Interest Rates	Aggressive Management's Most Likely Action:	Best Interest-Sensitive GAP Position to Be in:
Rates will raise	Reduce D_A and increase D_L (moving closer to a negative duration gap)	Net worth increase (if management's rate forecast is correct)
Rates will fall	Increase D_A and reduce D_L (moving closer to a positive duration gap)	Net worth increase (if management's rate forecast is correct)

7 Limitation of Duration Gap management

- Leverage-adjusted duration gap equals to zero is difficult:
  - Hard to find assets and liabilities of the same duration that fit into a financial-service institution's portfolio.
- Neglect convexity impact:
  - Only consider linear relationship
  - Major changes in interest rate does not apply
    - ∗ The accuracy and effectiveness of duration gap management decreases when there are major changes in interest rates.

考点 5 Illiquid Asset Ch19

1 Characteristics of illiquid markets

- Illiquidity: infrequent trading, small amounts being traded, low turnover
- Most asset classes are illiquid:
  - Equity (OTC): a week or more
  - Municipal bond: annual turnover of less than 10
  - Real estate: 4-5 years
  - Institutional infrastructure: 50 years or longer

  - Works of art: 40-70 years
- Markets for Illiquid Assets Are Large: the total wealth held in illiquid assets exceeds the total wealth in traditional, liquid stock, and bond markets.
  - U.S. residential mortgage market (2012): $16 trillion
  - Market capitalization of the NYSE and NASDAQ combined: $17 trillion
- Investor Holdings of Illiquid Assets: for individual investors, 90% total wealth is illiquid asset.
  - Human capital is the largest and least liquid asset for many individual investors.
  - High net worth individuals always hold 10-20% in treasure assets.
  - University endowments, pension funds, and other institutional investor also allocate much more on this asset kind.
- Liquidity Can Dry Up: In stressed economic periods, liquidity can dry up.
  - Illiquidity crises occur regularly because liquidity tends to dry up during periods of severe market distress.

2 Illiquid asset markets

- Source of illiquidity: illiquidity arises due to market imperfections
  - participation costs: investors must spend money, time, or energy to learn and gain necessary skills (capital, expertise, experience).
  - Transaction costs: commissions, taxes, due diligence, title transfer, etc.
  - Search frictions: need to search to find an appropriate buyer or seller.
  - Asymmetric information: one investor has superior knowledge compared with others so that other investors become reluctant to trade because concerning about fraud.
  - Price impact: large trades will move markets.
  - Funding constraints: access to credit is impaired

3 Biases on reported returns

- Survivorship bias: results from the tendency of poorly performing funds to stop reporting.
  - True illiquid asset returns are worse than the reported data.
- Reporting bias: occurs when fund never achieves a sufficiently attractive returns, they don't start reporting returns in the first place.
  - Overestimate the expected return
- Infrequent trading:
  - Return volatilities are underestimated.
  - Correlations with market (other assets) and betas are underestimated.
  - Returns for these infrequently traded assets are smoothed.

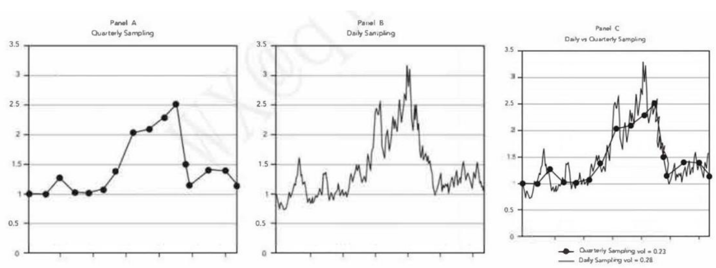


fig 15: sampling bias

- Selection bias: Asset values and returns tend to be reported when they are high. Sample selection bias results in overestimated expected returns and underestimated risk as measured by beta and the standard deviation of returns.
  - Buildings: many sellers postpone sales until property values recover.
  - Private equity: The venture capitalist tends to sell a small company, when the company's value is high and the transaction is recorded.

Biases	Results in
Survivorship biasReporting bias	Overestimated expected returns
Infrequent trading → smoothed data	Underestimated standard deviation Underestimated correlation with market (beta) Overestimated return autocorrelation
Selection bias	Overestimated expected returns Underestimated risk (σ, beta)

4 Illiquidity risk premiums

- Illiquidity risk premiums: compensate investors for the inability to access capital immediately.
  - Allocation across asset classes (e.g. real estate)
  - Choose securities within an asset class that are more illiquid
  - Act as market maker at the individual security level
  - Rebalancing

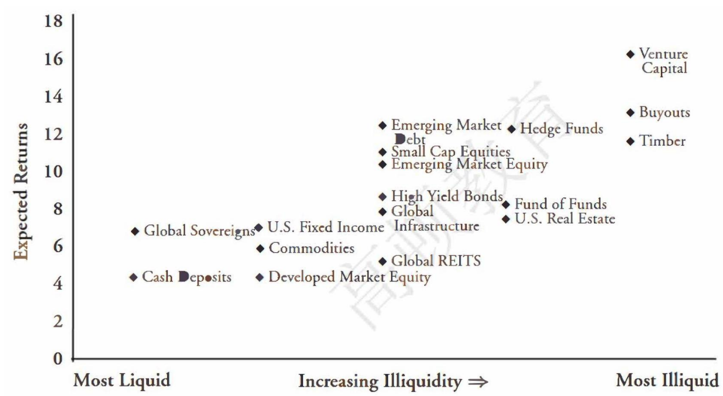


fig 16: liquidreturn

5 Illiquidity risk premiums across asset classes

- Conventional view: there is a reward to bearing illiquidity across asset classes. It may not be possible to generate illiquidity risk premiums across illiquid asset classes.
  - Illiquidity biases: reported data on illiquid assets cannot be trusted (survivorship, infrequent trading, selection bias).
  - Ignores risk: illiquid asset classes contain far more than just illiquidity risk (hedge fund and private equity fund). After adjusting for risk, these illiquid asset classes will be less compelling.
- There is an evidence of large illiquidity risk premiums within asset classes.
  - Within asset classes: take long positions in illiquid assets and short positions in liquid ones.
    - ∗ U.S. Treasuries: on-the-run/off-the-run
    - ∗ Corporate Bond: Larger bid-ask spreads and infrequent trading led to higher yields in corporate bond markets.
    - ∗ Equity: many illiquidity variables predict returns in equity markets, with less liquid stocks having higher returns.

- Why illiquidity risk premiums manifest within but not across asset classes?
  - Limited integration across asset classes: investors put asset classes into different silos and rarely treat them as a whole.

6 Liquidity providing activities

- Market making: A market maker supplies liquidity by acting as an intermediary between buyers and sellers.
  - Liquidity provision is costly: Investors transacting with the market maker pay the bid-ask spread.
- Rebalancing: the simplest way to provide liquidity, as well as the foundation of all long-horizon strategies.
  - Rebalancing forces asset owners to buy at low prices when others want to sell.
  - Since rebalancing is counter-cyclical, it supplies liquidity.

7 Portfolio choice with illiquid assets

- Investors face many considerations on how much of their portfolios to devote to illiquid assets:
  - Investors have different horizons
  - Illiquid market don't have tradable indices
  - Talented active portfolio managers are needed (agency issues)
  - How to evaluate and monitor portfolio managers
- Practice
  - For a portfolio of illiquid assets, hedge fund managers often have considerable discretion in portfolio valuation at the end of each month and may have incentives to smooth returns by marking values below actual in high-return months and above actual in low-return months. Which of the following is not a consequence of return smoothing over time?
    A. Higher Sharpe ratio
    B. Lower volatility
    C. Higher serial correlation
    D. Higher market beta
  - Answer: D
  - 解析: 根据题意，可以得知数据出现被平滑化，被平滑的数据会出现，波动率被低估，自相关系数被高估，市场 B 被低估的效果。因此，A，B，C 三个选项是平滑数据导致的结果不符合题意，D 选项符合题意，为正确选项。

3 Liquidity Management Strategies And Modeling

考点 1 Liquidity And Reserves Management: Strategies And Policies Ch5

1 The demand - supply framework

- Five sources of fund demand
  - Customers withdrawing money from their accounts
  - Credit requests from customers
  - Repayment of borrowings
  - Other operating expenses
  - Dividend payment to shareholders
- Five sources of fund supply

- New customer deposits: the most important source
- Customers repaying their loans
- Sales of assets
- Selling non deposit services
- Borrowings in the money market

• Net liquidity position (L_t)
  - L_t = Supplies of liquidity - Demands of liquidity
    * $L_t < 0$: Liquidity deficit
    * $L_t > 0$: Liquidity surplus

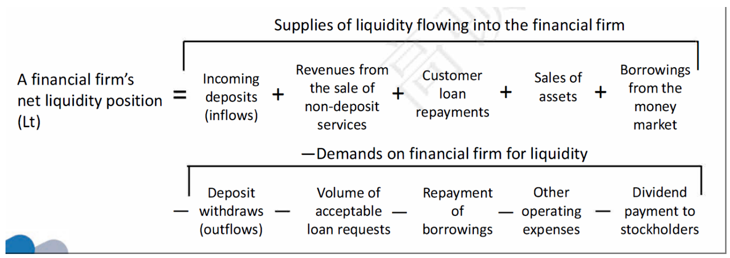


fig 17: net

• Examples
  - Over the next 24 hours, ABC Bank estimates that the following cash inflows and outflows (all figures in millions) will occur. What is the bank's projected net liquidity position?

Deposit withdrawals	\$35
Deposit inflows	\$20
Scheduled loan repayments	\$50
Acceptable loan requests	\$60
Borrowing from the money market	\$38
Sales of bank assets	\$90
Stockholder dividend payments	\$10
Revenues from sale of non - deposit services	\$22
Repayment of bank borrowings	\$30
Operating expenses	\$15

  - Answer:
    * Supplies of liquidity= 20 + 50 + 38 + 90 + 22 = 220
    * Demands of liquidity= 35 + 60 + 10 + 30 + 15 = 150
    * Net liquidity position = Supplies of liquidity - Demands of liquidity = 220 - 150 = 70 million dollars.

2 Strategies for liquidity management

• **Asset liquidity management strategies (Asset conversion strategies)**
  - **Definition:** Store liquidity in assets, mainly cash and marketable securities.
  - **Main users:** Smaller financial institutions.
  - **Pros:** Less risky than relying on borrowings.
  - **Cons:** High opportunity cost, as investing in liquid assets means forgoing higher returns on other assets.

• **Borrowed liquidity management strategies (Liability management strategies / Purchased liquidity strategies)**

- **Definition:** Raise liquid funds through borrowings in the money market.
- **Purest form:** Borrow immediately spendable funds to cover all anticipated demands for liquidity.
- **Main users:** Largest banks.
- **Pros:**
  * Lowering opportunity cost.
  * Leave assets portfolio unchanged, higher expected return.
  * Offer different rates to meet different funds needed.
- **Cons:** Most risky approach due to volatility of interest rates and rapid changes in credit availability.

• **Balanced liquidity management strategies**
  - **Definition:** expected liquidity demands are stored in assets (marketable securities) and advance arrangements for lines of credit. Unexpected cash needs are met by near-term borrowings.
  - **Pros:Strike a balance between risk and opportunity cost.**

Estimating liquidity needs

• :Sources and uses of funds approach
  Estimated liquidity deficit(-) or surplus (+) for the coming period:
  = Estimated change in deposits − Estimated change in loans

• Estimating future deposits and loans divides the forecast of future deposit and loan growth into three components:
  - A trend component
  - A seasonal component
  - A cyclical component

• **Example**
  - Forecasting deposits and loans with the sources and uses of funds approach (figures in millions of dollars)

Deposit Forecast for Deposits	Trend Estimate for Deposits	Seasonal Element	Cyclical Elements	Estimated Total Deposits
January, Week 1	$1,210	-4	-6	$1,200
January, Week 2	1,212	-54	-58	1,100
January, Week 3	1,214	-121	-93	1,000
January, Week 4	1,216	-165	-101	950
February, Week 1	1,218	+70	-38	1,250
February, Week 2	1,220	+32	-52	1,200

  - Forecasting deposits and loans with the sources and uses of funds approach (figures in millions of dollars)

Loan Forecast for Loans	Trend Estimate for Loans	Seasonal Element	Cyclical Elements	Estimated Total Loans
January, Week 1	$799	+6	-5	$800
January, Week 2	800	+59	-9	850
January, Week 3	801	+174	-25	950
January, Week 4	802	+166	+32	1,000
February, Week 1	803	+27	-80	750
February, Week 2	804	+98	-2	900

  - Forecasting liquidity deficits and surpluses with the sources and uses of funds approach (figures in millions of dollars)

Time Period	Estimated Total Deposits	Estimated Total Loans	Estimated Deposit Change	Estimated Loan Change	Estimated Liquidity Deficit (-) or Surplus (+)
January, Week 1	$1,200	$800			
January, Week 2	1,100	850	-100	+50	-150
January, Week 3	1,000	950	-100	+100	-200
January, Week 4	950	1,000	-50	+50	-100
February, Week 1	1,250	750	+300	-250	+550
February, Week 2	1,200	900	-50	+150	-200

3 Estimating liquidity needs

- The structure of funds approach
  - Total liquidity requirement = Deposit and nondeposit liability liquidity requirement + loan liquidity requirement
  - Deposit and non-deposit liability liquidity requirement: deposits and other funds are divided into categories based upon their estimated probability of being withdrawn:
    * ”Hot Money” liabilities (volatile liabilities)
    * Vulnerable funds
    * Stable funds (core deposits or core liabilities)
    * Liability liquidity reserve calculation: $0.95\times$ (Hot money deposits and nondeposit funds - Required legal reserve) $+0.30\times$ (Vulnerable deposit and nondeposit funds - Required legal reserve) $+0.15\times$ (Stable deposits and nondeposit funds - Required legal reserve)
  - Loan liquidity requirement:
    * 100% satisfy customers needs.
    * loan liquidity requirement $= 1.00\times$ (Potential loans outstanding - Actual loans outstanding)

4 Example: The structure of funds approach

- New Bank finds that its current deposits and non-deposit liabilities break down as follows:
  - Hot money: 30 million (95% liquidity reserve portion)
  - Vulnerable funds: 25 million (30% liquidity reserve portion)
  - Stable (core) funds: 120 million (15% liquidity reserve portion)
- Each kind of deposit has a required 3% legal reserve by the central bank. The bank’s total loans requirement recently are likely to be $160 million from $145 million actual loans. This financial firm decides to be ready to honor customer demands for all loans that meet its quality standards. What is the bank’s total liquidity requirement?
- **Answer:**

$$
\begin{aligned}
&\text{Total liquidity requirement calculation} \\
=&0.95 \times (30 - 0.03 \times 30) + 0.30 \times (25 - 0.03 \times 25) + 0.15 \times (120 - 0.03 \times 120) + 100\% \times (160 - 145) \\
=&67.38 \text{ million}
\end{aligned}
$$

5 Estimating liquidity needs

- The structure of funds approach

- Refine the structure of funds approach by calculating expected liquidity requirements based on probabilities of different possible outcomes.

$$\begin{aligned}\text{Expected liquidity requirement} &= \text{Probability of Outcome A} \times (\text{Estimated liquidity surplus or deficit in Outcome A}) \\ &+ \text{Probability of Outcome B} \times (\text{Estimated liquidity surplus or deficit in Outcome B}) \\ &+ \cdots + \cdots\end{aligned}$$

6 Example: refine the structure of funds approach

	Estimated Deposits Next Week (millions)	Estimated Loans Next Week (millions)	Estimated Liquidity Position Next Week (millions)	Probability to Each Possible Outcome
Best position	190	100	+90	15%
Normal position	170	130	+40	65%
Worst position	140	160	-20	20%

Expected liquidity requirement = 90 × 15% + 40 × 65% + (-20) × 20% = 35.5 million

7 Estimating liquidity needs

- Liquidity indicator approach
  - Many financial-service institutions estimate their liquidity needs based upon experience and industry averages, which means using certain liquidity indicators.

Liquidity indicator	Description	Liquidity impact (the higher the ratio)
Cash position indicator	Cash and deposits from depository institutions / Total assets	The smaller the chance of a liquidity crisis
Core deposit ratio	Core deposits / Total assets	The smaller the chance of a liquidity crisis
Deposit composition ratio	Demand deposits / Time deposits	The greater the chance of a liquidity crisis
Deposit brokerage index	Brokered deposits / Total deposits	The greater the chance of a liquidity crisis
Liquid securities indicator	U.S. gov. securities / Total assets	The smaller the chance of a liquidity crisis
Pledged securities ratio	Pledged securities / Total securities holdings	The greater the chance of a liquidity crisis
Hot money ratio	Money□market assets / Volatile liabilities	The smaller the chance of a liquidity crisis
Capacity ratio	Net loans & leases / Total assets	The greater the chance of a liquidity crisis
Loan commitments ratio	Unused loan commitments / Total assets	The greater the chance of a liquidity crisis
Net Fed funds & repo position	(Fed funds sold + rev. repos − Fed funds purchased − repos) / Total assets	The smaller the chance of a liquidity crisis

table 4: Comprehensive Liquidity Indicators

8 Legal reserve

- Lagged reserve accounting (LRA) is used to calculate Legal Reserves.

- Reserve computation period: two-week period from a Tuesday through a Monday two weeks later.
- Reserve maintenance period: 14-day period stretching from a Thursday through a Wednesday.
- Reserve funding period: reserve maintenance begins 30 days after the beginning of the reserve computation, which allows the money position manager 16 days to plan.

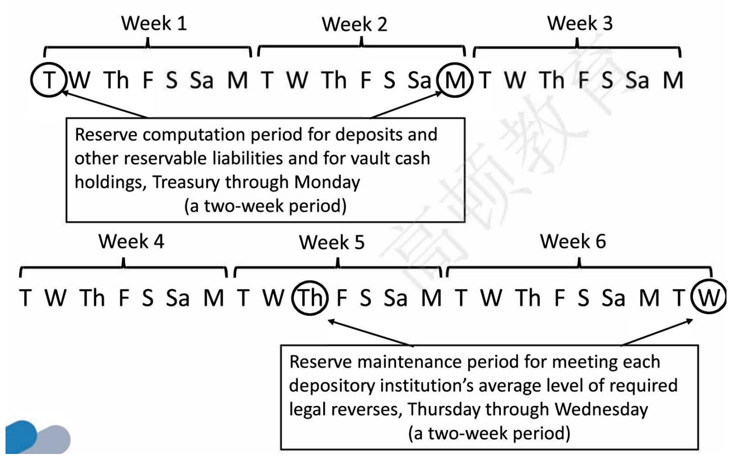


fig 18: Legal Reserves

- Total required legal reserves = Reserve requirement on transaction deposits × Daily average amount of net transaction deposits over the computation period + Reserve requirement on nontransaction reservable liabilities × Daily average amount of nontransaction reservable liabilities over the computation period
- Alternative sources of reserves:
  - Federal funds market
  - Selling liquid securities
  - Drawing upon any excess correspondent balances placed with other depository institutions
  - Repo
  - Selling new time deposits to the largest customers
  - Borrowing in the Eurocurrency market
  - Central bank's discount window
- Factors in choosing among the different sources :
  - Immediacy of need
  - Duration of need
  - Access to the market for liquid funds
  - Relative costs and risks of alternative sources of funds
  - The interest rate outlook
  - Outlook for central bank monetary policy
  - Rules and regulations applicable to a liquidity source

考点 2　Monitoring Liquidity Ch7

1　Liquidity options

- Definition: give the right of a holder to receive cash from or to give cash to the bank at predefined times and terms.

2 Liquidity options Examples:

- **Credit line sold to customer:** the obliger has the right to withdraw whatever amount up to the notional of the line.
  - Customer's reasons for exercised
    - ∗ Provide beneficial financing rate
    - ∗ Generate cash flow conveniently
- **Sight and saving deposits:** the bank's clients can typically withdraw all or part of the deposited amount with no or short notice.
  - Customer's reasons of exercise
    - ∗ Invest in assets with higher yields
    - ∗ Liquidity need for transaction purposes
- **Prepayment of fixed rate mortgages:** the bank's client has the right to repay the funds before the contract end.
  - Customer's reasons of exercise:
    - ∗ Divorces or retirements
    - ∗ Financial incentive to close the contract and reopen it under current market conditions.

3 Impact of liquidity options

- Liquidity impact on the balance sheet: amount withdrawn or repaid.
  - Negative impact: credit lines, sight/saving deposits.
  - Positive impact: prepayment of fixed rate mortgages.
- Financial impact: difference between contract and market rates at exercise.

4 A taxonomy of cash flows

- The taxonomy focuses on two main dimensions: time and amount. We can further classify these two dimensions as deterministic and stochastic.
- Example

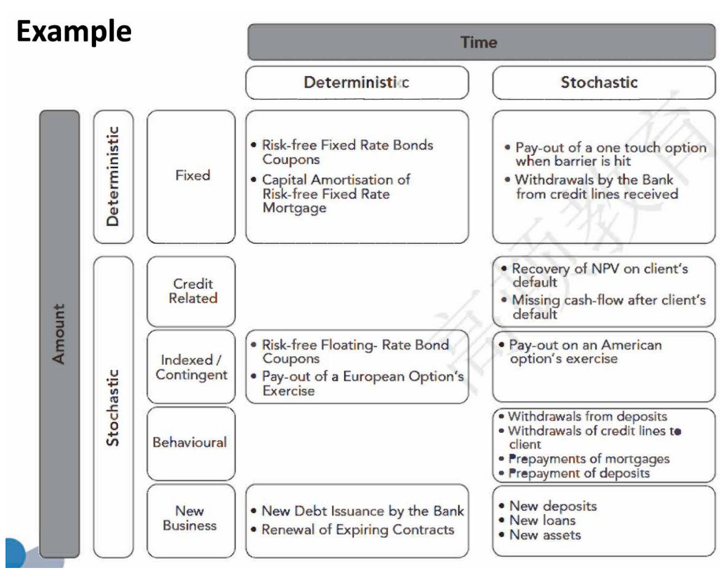


fig 19: taxonomy

5 Term structure of expected cash flows (TSECF):

- Definition: TSECF can be defined as the collection, ordered by date, of expected cash flows, up to expiry referring to the contract with the longest maturity.

6 Term structure of expected cumulated cash flows (TSECCF):

- Definition: TSECCF is the collection of expected cumulated cash flows ordered by date:
  - Represents how the past dynamic evolution of net cash flows affects its total cash position on that date.

7 Example: TSECF & TSECCF

- Consider a bank with a simplified balance sheet below:

Asset		Liability	
bonds	30	bonds	40
loans	70	deposits	40
		Equity	
		equity	20

- One key step to build the TSECF is to order the assets and the liabilities according to their maturity.

Time	Asset cf_e^+	Liabilities cf_e^-
1	20	
2		-10
3		
4		
5	50	
6		
7		-70
8		
9		
10	30	
> 10		-20

- Table below shows a very simple balance sheet producing a very simple TSECF and related TSECCF.

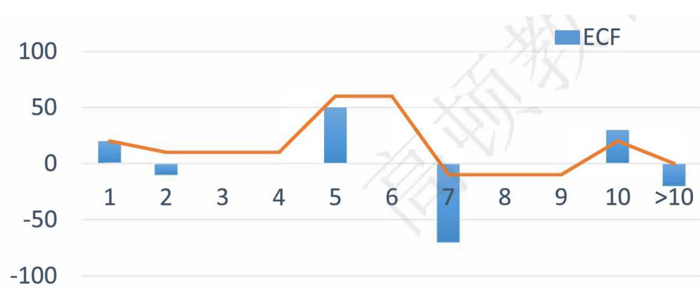


fig 20: ECF

- Negative values in TESCCF: on an expected basis, this means the bank may become insolvent and eventually go bankrupt.

8 Liquidity generation capacity (LGC)

- Definition: LGC is the main tool a bank can use to handle the negative entries of the TSECCF.
  - The ability of a bank to generate positive cash flows, beyond contractual ones, from the sources of liquidity available in/off the balance sheet at a given date.
- Three types of sources of liquidity:
  - Selling of Assets (AS)
  - Secured funding using assets as collateral
    - ∗ Repo transactions (RP)
  - Unsecured funding (USF)
    - ∗ Withdrawals of committed credit lines available from other financial institutions
    - ∗ Deposit transactions in the interbank market

9 The term structure of LGC (TSLGC)

- Definition: TSLGC as the collection of liquidity that can be generated at a given time, by the sources of liquidity, at the reference time up to a terminal time.

10 Example: TSLGC & TECLGC

Time	AS	RP	USF	LGC	CLGC
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	0	0	0	0	0
5	0	0	0	0	0
6	0	0	0	0	0
7	2	5	3	10	10
8	1	2	1	4	14
9	-0.5	-1.5	-0.5	-2.5	11.5
10	-2.5	-5.5	-3.5	-11.5	0
> 10	0	0	0	0	0

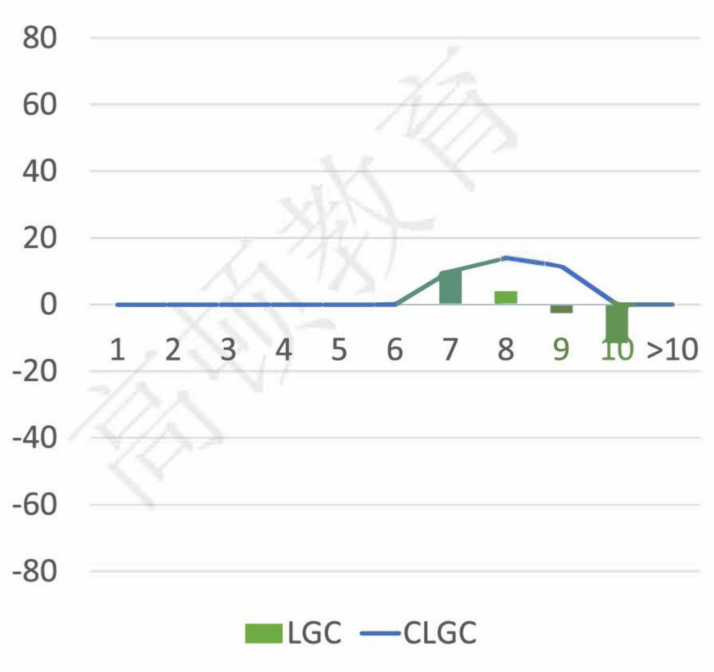


fig 21: fig 21: LGC

11 The term structure of expected liquidity (TSL_e)

- Definition: TSL_e is basically a combination of the TSECCF and the TSCLGC.
  - TSL_e is in practice a measure to check whether the financial institution is able to cover negative cumulated cash flows at any time in the future, calculated at the reference date.
  - TSL_e represents the main monitoring tool of a Treasury Department.

12 Example: TSL_e

Time	TSECCF	TSCLGC	TSLe
1	20	0	20
2	10	0	10
3	10	0	10
4	10	0	10
5	60	0	60
6	60	0	60
7	-10	10	0
8	-10	14	4
9	-10	11.5	1.5
10	20	0	20
> 10	0	0	0

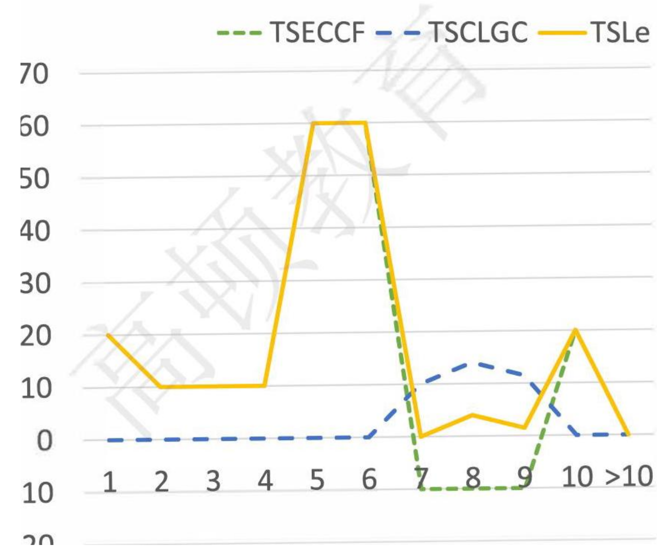


fig 22: fig 21: TSL

13 The term structure of available assets (TSAA)

- Definition: TSAA is a tool used to monitor a bank's LGC and to provide an indication of the extent to which an asset can be used to create liquidity.
  - Only shows how many single securities are available for inclusion in LGC.
    - Does not imply that the notional amount of securities can be fully converted into liquidity.

14 Transactions affect TESCF/TSAA/TSLGC

table 5: Transaction Types and Their Impact on Liquidity Monitoring

Type	Ownership	Possession	Changes to TSECF	TSAA	TSLGC
Buy	Bank	Bank	Yes	+	Possible
Sell	Others	Others	Yes	-	Yes
Repo	Bank	Others	No	-	Yes
Reverse Repo	Others	Bank	Yes	+	Possible
Sell/Buyback	Others	Others	Yes	-	Yes
Buy/Sellback	Bank	Bank	Yes	+	Possible
Security Lending	Bank	Others	Yes	-	No
Security Borrowing	Others	Bank	Yes	+	Possible

15 Cash flows at risk (cfaR):

- Cash flows are stochastic: not only the expected value but also its volatility.
- Cash flows at risk: showing the extreme values that both positive and negative cash flows assume during the time of their occurrence.
- Positive cash flow at risk :

$$\text{cfaR}^{+}_{\alpha}(t_0, t_i) = \text{cf}^{+}_{\alpha}(t_0, t_i; x) - \mathbb{E}[\text{cf}(t_0, t_i; x)]$$

- Negative cash flow at risk :

$$\mathrm{cfaR}^{-}_{1-\alpha}(t_0,t_i) = \mathrm{cf}^{-}_{1-\alpha}(t_0,t_i;x) - \mathbb{E}[\mathrm{cf}(t_0,t_i;x)]$$

  – x: risk factors such as credit, behavioral and market variables e.g. interest rate
  – α: confidence level e.g. $\alpha = 99\%$

- Term structure of unexpected positive cash flows:

$$\mathrm{TSCF}^{+}_{\alpha}(t_0,t_b) = \left\{\mathrm{cfaR}^{+}_{\alpha}(t_0,t_0), \mathrm{cfaR}^{+}_{\alpha}(t_0,t_1), \dots \mathrm{cfaR}^{+}_{\alpha}(t_0,t_b)\right\}$$

  – Upper bound of TESCF

- Term structure of unexpected negative cash flows:

$$\mathrm{TSCF}^{-}_{1-\alpha}(t_0,t_b) = \left\{\mathrm{cfaR}^{-}_{1-\alpha}(t_0,t_0), \mathrm{cfaR}^{-}_{1-\alpha}(t_0,t_1), \dots \mathrm{cfaR}^{-}_{1-\alpha}(t_0,t_b)\right\}$$

  – Lower bound of TESCF

- Example: Maximum, minimum ($\alpha = 99\%$) and expected cash flows for a period of 20 days

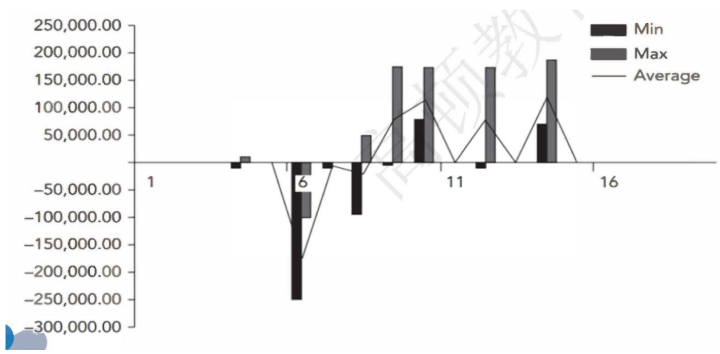


fig 23: CFAR

16 Monitoring liquidity framework summary

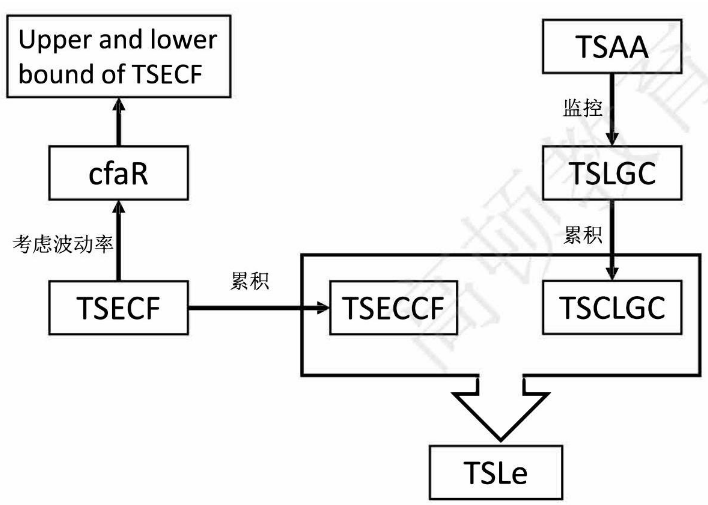


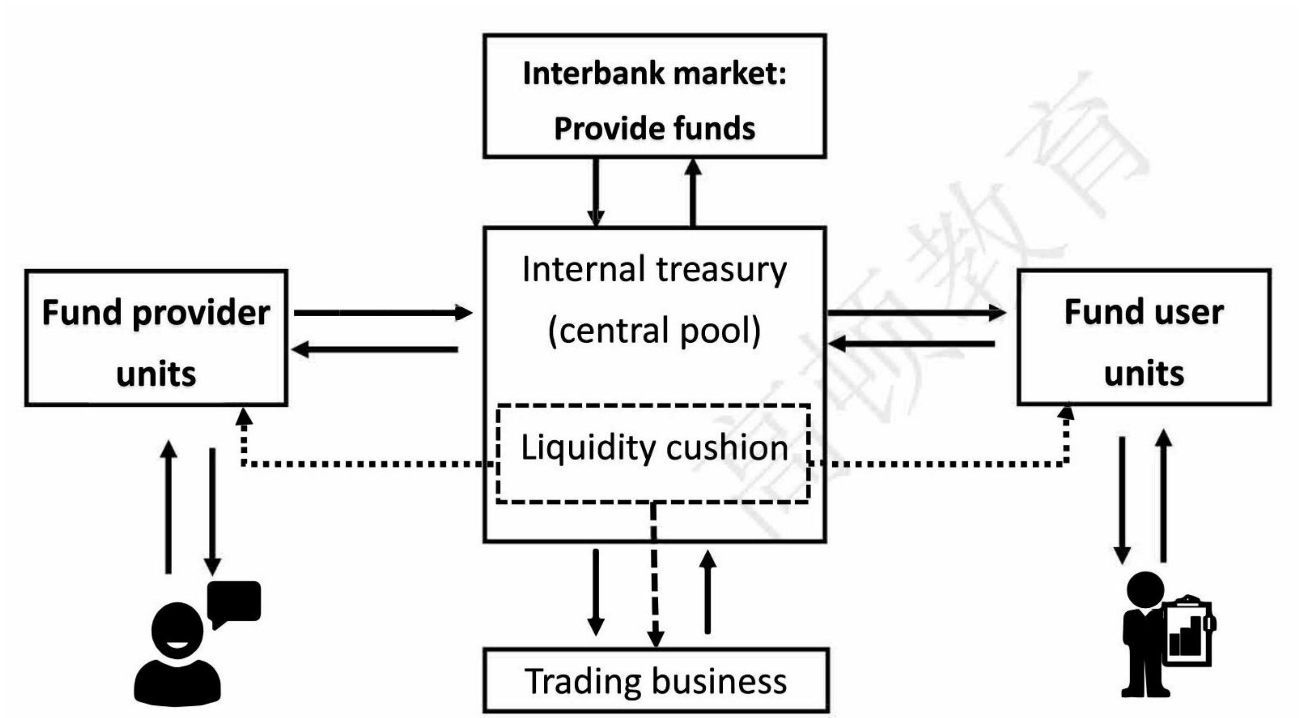
fig 24: TSLe

考点 3 Liquidity Transfer Pricing: A Guide Better Practice Ch15

1 Introduction of LTP

- Liquidity Transfer Pricing (LTP) is a process that attributes the costs, benefits, and risks of liquidity to respective business units within a bank.
  - Purpose of LTP:
    - Transfer liquidity costs and benefits from business units to a centrally managed pool.
    - Recoup the cost of carrying a liquidity cushion by charging contingent commitments.

2 Process of LTP



3 Best Practices for LTP Process

- Governance of the LTP process
- Best practice 1: establish adequate LTP policies and procedures.
- Best practice 2: better to have a centralized funding center. Wholesale funding should be confined to this function.
- Best practice 3: develop trading book policies and identify funding requirements.
- Best practice 4: set up an effective oversight of the LTP process by independent risk and financial control personnel.
- Best practice 5: improve risk-adjusted profit measures to prompt business units to consider the cost of liquidity as part of their decision to book certain assets.
- Best practice 6: related parties involved in the management of LTP should better understand the LTP process.

4 Challenges in LTP Implementation

- Difficult to maintain proper oversight to manage and monitor the LTP process.
- The Liquidity Management Information Systems (LMIS) are unable to attribute the costs, benefits, and risks of liquidity appropriately to respective businesses at a sufficiently granular level.

– LMIS are widely used by management as a primary source of measuring and monitoring the performance of businesses.

5 Approaches of LTP

- **Zero cost of funds:**
  - Definition: View funding liquidity as free, and funding liquidity risk as zero. (in an easy funding condition)
    - Assume interest rate risk is properly accounted for using the swap curve: a zero spread above the swap curve implies a zero charge for the cost of funding liquidity.
  - Evaluation: the worst practice
  - Results: the hoarding of long-term highly illiquid assets
    - Few long-term stable liabilities to meet funding demand.

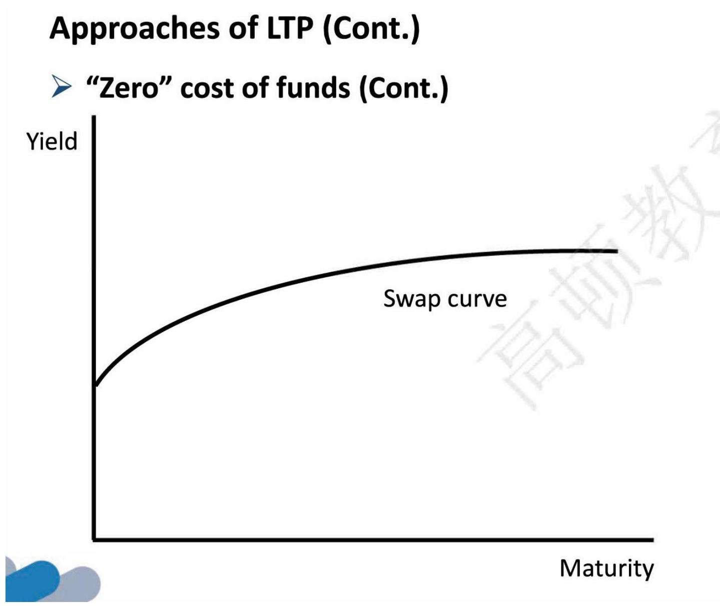


fig 25: zero

- **Average cost of funds:**
  - Pooled "average" cost of funds
    - Definition: All assets and liabilities irrespective of their maturity are charged the same rate for cost/benefits of liquidity.
      - Average rate calculated are based on all existing funding sources.
    - Evaluation: better than zero cost of funds

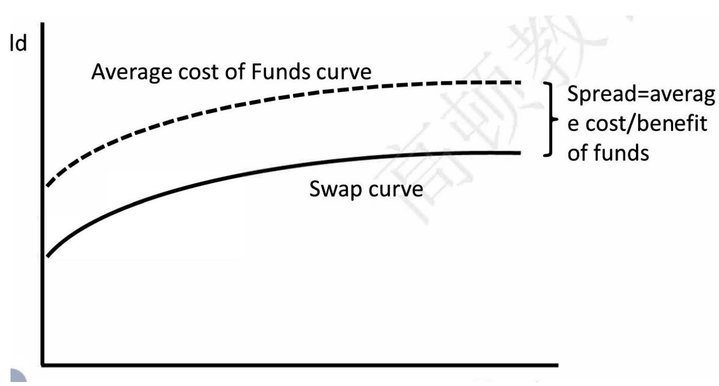


fig 26: Pooled

Terms in years	1	2	3	4
Loans/deposit principal	1 million	1 million	1 million	1 million
Average cost of funds (bps)	8	8	8	8
Charges for use of fund	800	800	800	800
Credits for benefit of fund	800	800	800	800

table 6: Loan and Deposit - related Data

- Separate ”average” rates for cost and benefits of funds
  - Definition: average rates for cost of funds and benefits of fund are calculated separately.
    - Loans can be encouraged or discouraged by adjusting the average cost of using funds, without affecting the solicitation of funds by fund providers.
  - Evaluation: better than pooled ”average” cost of funds

Terms in years	1	2	3	4
Loans/deposit principal	1 million	1 million	1 million	1 million
Average cost of funds (bps)	8	8	8	8
Average benefit of funds (bps)	3	3	3	3
Charges for use of fund	800	800	800	800
Credits for benefit of fund	300	300	300	300

table 7: Financial Terms and Values

- Results:
  - Neglects the varying maturity of assets and liabilities by applying a single charge for the use and benefit of funds
  - Lags changes in banks’ actual market cost of funding.
  - Promotes unhealthy maturity mismatch
  - Distorts profit assessment

- **Matched-maturity marginal cost of funds:**
  - Definition: using term liquidity premium to convert fixedrate borrowing costs to floating-rate borrowing costs through an internal swap transaction.
  - Evaluation: current best practice but more complex than average cost

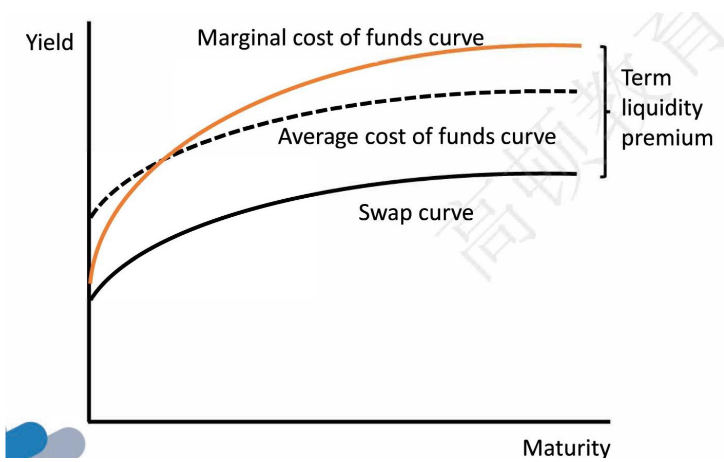


fig 27: Matched

  - Results:

Terms in years	1	2	3	4
Term liquidity premium	5	10	18	28
Average cost of funds (bps)	8	8	8	8

table 8: Liquidity and Cost Data

    * Recognize the costs and benefits of liquidity are related to the maturities of assets and liabilities.
        · Assign higher rates to products that use or provide longer-term liquidity.
    * Recognize the importance of having changes in market conditions incorporated into the rate used to charge and credit users and providers of funds. (actual market cost)

6 Contingent Liquidity Risk Pricing

- Contingent commitments may result in contingent liquidity risk.
  - Lines of credit, collateral postings for derivatives and other financial contracts, etc.
- Liquidity cushion: Banks carry a liquidity cushion, a buffer of highly liquid assets or stand-by liquidity to survive periods of unexpected funding outflows.
  - Use the results of stress-testing to determine the size of the liquidity cushion.

7 Contingent liquidity risk pricing process

- Process of Contingent liquidity risk pricing
  - Step 1: Identify contingent commitments that are likely to create unexpected funding demands.
  - Step 2: Perform stress tests under various scenarios to approximate the funding that might be required.
  - Step 3: Net approximations from above against inflows generated.
    * For example: through the sale of marketable securities to derive the size of the liquidity cushion.
  - Step 4: Calculate the cost of carry: the cost of funding liquid assets minus the return they generate.
  - Step 5: Recoup cost of carry: charging a liquidity premium, at the most granular level, to the business unit, product or transaction that creates the need for the bank to carry such liquid assets.

8 Contingent liquidity risk pricing calculation

- Method for pricing contingent liquidity risk in credit lines
  - At the most basic level of what is considered to be better practice, all banks should be charging contingent commitments based on their likelihood of drawdown.
  - Rate charge formular:

$$\frac{\text{limit} - \text{drawn amount}}{\text{limit}} \times \text{likelihood of drawdown} \times \text{cost funding liquidity cushion}$$

  - Example
    * Suppose a line of credit with a limit of \$10 million has \$4 million already drawn. A 60 percent chance the customer will draw on the remaining credit and the cost of term funding assets in the liquidity cushion is 18 bps. Please calculate the rate cost of contingent liquidity risk.
  - Answer

$$\text{Dollar charge} = (\$10m - \$4m) \times 0.6 \times 0.0018 = 6480$$

$$\text{Rate charge} = \frac{6480}{\$10m} = 6.48\text{bps}$$

考点 4 US dollar Shortage and Covered Interest Parity lost Ch 16 & 17

1 Covered Interest Parity

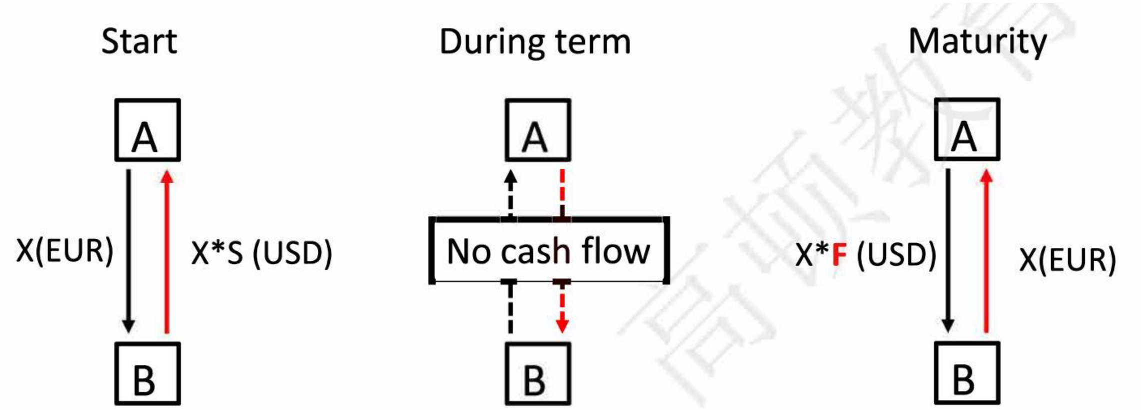
- CIP is a no-arbitrage condition according to which interest rates on two otherwise identical assets in two different currencies should be equal once the foreign currency risk is hedged:

$$\frac{F}{S} = \frac{1+r}{1+r^*}$$

  - S: spot exchange rate in units of US dollar per foreign currency
  - F: corresponding forward exchange rate
  - r: the US dollar interest rate
  - r^*: the foreign currency interest rate.
- In practice, the relationship between F and S is observed in market transactions in FX instruments, notably FX swaps and cross-currency swaps.

2 FX Swaps vs. Cross-currency Swaps

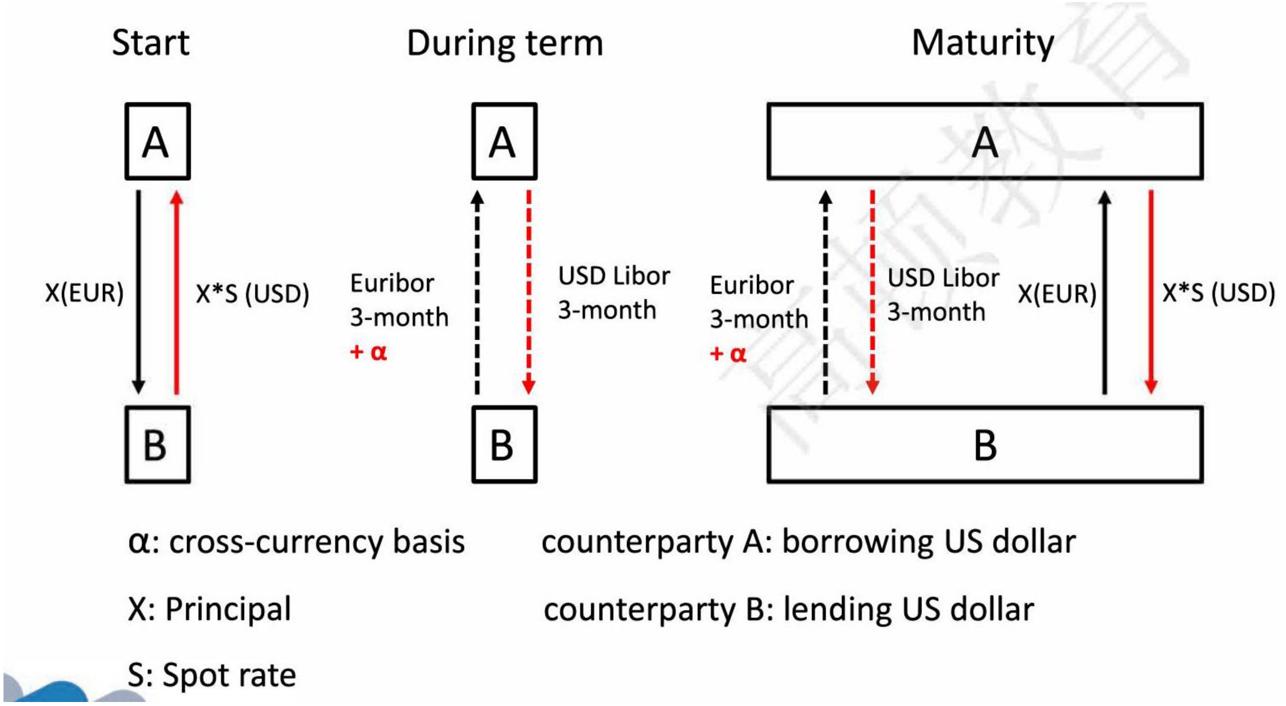
- **Mechanics of FX swaps:**
  - See diagram below.
  - S: Spot exchange rate; F: predetermined forward exchange rate; X: Principal.
  - Counterparty A: borrowing US dollar; Counterparty B: lending US dollar.



  - If $b \neq 0$, the CIP fails: the party lending a currency via FX swaps makes a higher or lower return than implied by the interest rate differential in the two currencies.

$$\frac{F}{S} = \frac{1+r^{USD}+b}{1+r^{foreign}}$$

- **Mechanics of cross-currency swaps:**



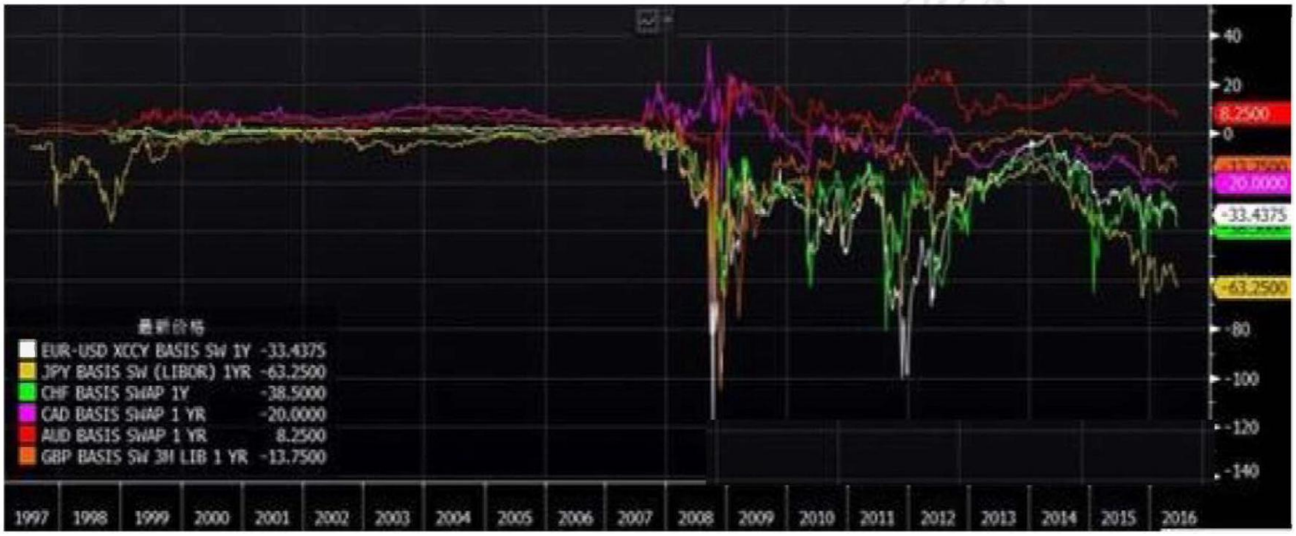
- When $\alpha \neq 0$, the failure of CIP if the party lending US dollars in a cross-currency swap pay the basis, non-zero α, on top of foreign currency.
- In a cross-currency basis swap, interest payments are based on the respective Libor rates plus the basis α.

3 Introduction of Cross-Currency Basis

- Key factors affecting the cross-currency swap basis:
  - Drop in market liquidity
    - Market liquidity in the underlying instruments may evaporate, so that the difference between bid and ask prices for forward and spot transactions is non-trivial.
  - Risk increases of the underlying instruments
    - For example, a rise in counterparty credit risks in the interbank markets

4 The violation of CIP

- Negative basis points period: Since 2007, the basis for lending US dollars against most currencies, notably the euro and yen, has been negative.
  - Borrowing dollars through the swap market became more expensive than direct funding in the dollar cash market.
- Reflection of CIP violation: strains in global interbank markets.
  - Counterparty risk
  - Constrained bank access to wholesale dollar funding inhibited arbitrage
- Figure of XCCY (α: cross-currency basis)



5 Causes Of CIP Violation

- Demand for currency hedges
  - The reason of why the basis opens up
- New constraints on arbitrage activity
  - The reason why the basis does not close

6 Demand for currency hedges

- Three sources of hedging demand
  - Demand from banks
    * Foreign currency hedges from banks' business models
  - Demand from institutional investors
    * Strategic hedging decisions of institutional investors
  - Demand from Non-financial firms
    * Non-financial firms' debt issuance across currencies

7 New constraints on arbitrage activity

- Market structural changes tighten limits to arbitrage.
  - Structural changes in how market participants have been pricing market, credit, counterparty and liquidity risks postcrisis have tightened limits to arbitrage.
- Balance sheet space is not free and arbitrage can be both costly and risky.
  - Arbitrage enlarges its balance sheet, and banks possibly face credit risk, mark-to-market and liquidity risk.
- Changes in regulation have reinforced market pressures for a tighter management of balance sheet risks.
  - Derivatives transaction changes: changes related to credit value adjustments have sought to incentivize dealers to price the counterparty risk more accurately
  - Basel Ill and US leverage ratios changes: potential future exposure adjustment charges require market participants to hold capital in proportion to their derivatives and other exposures.

8 Background introduction

- Introduction: Global banking activity had grown remarkably between 2000 and mid-2007. As bank's balance sheets expanded, they appetite for foreign currency assets like US dollar loans and structural products.
  - Banks borrow US dollars to invest:
    - Borrow domestic currency and convert to U.S. dollars in the spot market.
    - Using FX swap convert domestic currency to US dollars
    - Borrowing US dollar from interbank market.

9 Maturity/currency transformation

- US dollar maturity funding gap
  - US dollar investment to non-banks are funded by liabilities to various counterparties with various maturities, subjected to considerable funding risk.
    - The amount of US dollars invested in longer-term assets which is not supported by longer-term US dollars liabilities.
  - This gap must roll over before their investment mature.

10 Causes of the shortage of US dollars

- Unsustainable maturity transform: The major sources of short-term funding turned out to be less stable than expected.
  - FX swap became expensive: The heightened counterparty risk and liquidity concerns in FX swap markets made it even more expensive to obtain US dollars via currency swaps.
  - Non-bank sources of funds is instable: Money market funds, facing large redemptions following the failure of Lehman Brothers, withdrew from bank-issued paper, threatening a wholesale run on banks.
  - Market conditions during the crisis have made it difficult for banks to respond to funding pressures by reducing their US dollar assets: such as structured finance products, were harder to sell without realising large losses.

11 The international policy response

- Temporary reciprocal currency arrangement (swap lines) between Federal Reserve and European central banks: The Federal Reserve channel US dollars to European central banks, then the European central banks channel to European banks in their respective jurisdictions unlimitedly.
- Swap lines extended across continents to central banks in Australia and New Zealand, Scandinavia, and several countries in Asia and Latin America, forming a global network.
- The advantages of the swap line between central banks:
  - 央行具有货币发行权: Federal Reserve and its central bank counterparties have the power to create any amount of money, in contrast with international financial institutions administering limited resources.
  - 无信息不对称引起的道德风险: Swap network does not compound the informational problems that can give rise to moral hazard.

4 Liquidity Risk Management Tools

考点 1 Early Warning Indicators chap3

1 Definition

- Early warning indicators (EWls) are analogous to warning lights on a car dashboard.
- EWIs are changes in key metrics (qualitative or quantitative) that could signal a pending liquidity problem and they can vary in severity and priority.

2 EWI & Liquidity risk management process (LRM)

- LRM framework begins by identifying liquidity and funding risks that are inherent in the business activities.
- LRM framework should be end-to-end and ensure that the bank will ”sufficiently capture the banking organization's exposures, activities, and risks.”
- Ensure that EWls are updated and aligned with other core aspects of the LRM framework.
- Ensure the quality and timeliness of the data that feeds into the EWIs.

3 Framework Components: M.E.R.I.T

- Measures
  - Measures should be leading and sharp.
    - ∗ Leading(forward looking): provide information and signal potential stress prior to the occurrence of an actual event.
    - ∗ Sharp(sufficiently granular): signals do not go unnoticed within the mass of data.
  - Measures should contain both external and internal measures.
    - ∗ Internal measures: customize bank's B/S and activities.
    - ∗ External measures: signal systemic changes in the economy or market.
- Escalation
  - An effective escalation policy should ensure that limit breaches are escalated to the appropriate level of management with the authority to undertake corrective actions.
    - ∗ Modest response: increase liquid asset buffer
    - ∗ Large cash outflow from EWI: asset-liability committee (ALCO)
    - ∗ Extreme cases: contingency funding plan (CFP)
- Reporting
  - EWI report needs to be timely in order to provide management with sufficient lead time to make adjustment in response to potential crisis events. It is common practice to have the EWI dashboard reported on a daily basis.
- Integrated systems
  - Metrics and reporting are feasible and meaningful only when they are backed by the bank's ability to calculate the selected metrics and report them consistently.
- Thresholds
  - Firms use a stoplight system in represent and communicate performance against the thresholds of their EWls.
    - ∗ Green indicator: the measure is within normal bounds.
    - ∗ Amber indicator: the measure should be investigated further.

    * Red indicator: significant concern and may need an immediate response.
  - The threshold boundaries for which an EWI moves from green to amber should not be too wide or too narrow.
    * Use back-testing to determine if recalibration is needed.

4 Guidelines from OCC

- EWI may include but not limited to the items listed below:
  - A reluctance of traditional fund providers to continue funding at historic levels.
  - Pending regulatory action(both formal and informal) or CAMELS component or composite rating downgrade(s).
    * CAMELS: capital, asset quality, management, earnings, liquidity, and sensitivity.
  - Widening of spreads on debts, credit default swaps, and stock price declines.
  - Difficulty in accessing long-term debt markets.
  - Reluctance of trust managers, money managers, public entities, and credit-sensitive funds providers to place funds.
  - Rising funding costs in an otherwise-stable market.
  - Counterparty resistance to off-balance-sheet products or increased margin requirements.
  - The elimination of committed credit lines by counterparties.

5 Guidelines from BCBS

- EWI may include but not limited to the items listed below:
  - Rapid asset growth, especially when funded with potentially volatile liabilities.
  - Growing concentrations in assets or liabilities.
  - Increases in currency mismatches.
  - Decrease of weighted average maturity of liabilities.
  - Repeated incidents of positions approaching or breaching internal or regulatory limits.
  - Negative trends or heightened risk associated with a particular product line.
  - Negative publicity
  - Credit rating downgrade
  - Stock price declines
  - Rising debt costs
  - Widening debt/credit-default-swap spreads
  - Rising wholesale/retail funding costs
  - Counterparties requesting additional collateral or resisting entering into new transactions
  - Drop in credit lines
  - Increasing retail deposit outflows
  - Increasing redemptions of CDs before maturity
  - Difficulty accessing longer-term funding
  - Difficulty placing short-term liabilities
- Intraday liquidity monitoring indicators include:
  - Daily maximum liquidity requirement
  - Available intraday liquidity
  - Total payments

- Time-specific and other critical obligations
- Value of customer payments made on behalf of financial institutions customers
- Intraday credit lines extended to financial institution customers
- Timing of intraday payments
- Intraday throughput

6 Guidelines from Federal Reserve

- EWI may include but not limited to the items listed below:
  - Negative publicity concerning an asset class owned by the institution.
  - Increased potential for deterioration in the institution's financial condition.
  - Widening debt or credit default swap spreads.
  - Increased concerns over the funding of off-balance-sheet items

考点 2 Intraday Liquidity Risk Management Ch6

1 Basics of intradav liauiditv

- A bank treasurer manages intraday liquidity risk and enumerates the sources and uses of intraday funds for a typical large bank.
  - Target: Ensure the bank remains within the limits of its daylight credit resources and ends the day with the appropriate target balances in its accounts.
  - Challenges: factors of intraday funds management target
    * Variability of cash flow patterns
    * Impact of market forces
    * Lack of real-time data

2 Uses of intraday liquidity fund

- Outgoing Wire Transfers (largest use of intraday liquidity)

Description	✓Payments on LVPS such as Fedwire and CHIPS
Impacted by client activity	✓Yes ✓Clients provide bank instructions
Impacted by bank activity	✓Yes ✓Bank treasury and lines of business have payment needs
Ability to forecast	✓Bank activity can usually be forecasted with 1 to 2 days' notice ✓client payment activity is more difficult to predict

- Settlements at PCS systems

Description	✓Net cash settlements at payments systems, clearing houses
Impacted by Client Activity	✓Not directly initiated by clients, but includes client activity
Impacted by Bank Activity	✓Yes ✓Position monitored by operations groups
Ability to forecast	✓Can be forecast for securities that have multi-day settlements (e.g., T +3) ✓More difficult same day settlement activity

- Funding Nostro Accounts

Description	✓Cash transfer to a correspondent bank for services provided
Impacted by Client Activity	✓Not directly initiated by clients, but includes client activity
Impacted by Bank Activity	✓Yes ✓Position monitored by operations groups
Ability to forecast	✓Securities settlements are generally predictable ✓Client payment activity flowing through correspondents is less predictable

- Collateral Pledging

Description	✓Obtaining and earmarking of collateral required by an outside beneficiary
Impacted by Client Activity	✓Yes ✓some clients require collateral to cover bank trading liabilities or deposits
Impacted by Bank Activity	✓Yes ✓banks are required to post collateral for margin at FMUs other trading counterparties
Ability to forecast	✓Dependent on trading volumes and asset price changes, generally known one day in advance

- Asset Purchases/Funding

Description	✓Exchange of bank cash for another asset such as a client loan
Impacted by Client Activity	✓Yes ✓Clients can draw down on lines of credit or letters of credit
Impacted by Bank Activity	✓Yes ✓Assets may be securities for the bank’s investment portfolio or fixed assets
Ability to forecast	✓Bank fixed asset activity should be known in advance ✓Securities purchases may be same day settlement ✓Client loans are more difficult to predict

3 Sources of intraday liquidity fund

- Cash Balances
- Incoming Funds Flow
- lntraday credit
- Liquid Assets
- Overnight Borrowings
- Other Term Funding

Description	✓Deposits at central bank and correspondent bank nostro accounts
Impacted by Client Activity	✓No ✓clients do not directly impact closing/start of day cash balances
Impacted by Bank Activity	✓Yes ✓Bank Treasury determines level of closing/start of day cash balances
Ability to forecast	✓Bank activity can usually be forecasted with 1 to 2 days' notice ✓Client payment activity is more difficult to predict

Description	✓Incoming cash payments and cash credits from FMU
Impacted by Client Activity	✓Yes ✓Incoming client payments are credited to the bank's accounts at the central bank and correspo
Impacted by Bank Activity	✓Yes ✓Activities conducted for the bank's business can impact cash balances
Ability to forecast	✓Bank activity can usually be forecasted with 1 to 2 days' notice ✓Client payment activity is more difficult to predict. ✓FMU credit can be forecast for securities that have multi - day settlements (e.g., T + 3) and more difficul

Description	✓Credit line or overdraft permitted during business hours and covered by close of business. ✓Lines are often uncommitted and provided without interest charges.
Impacted by Client Activity	✓No, client activity does not directly impact the amount of intraday credit extended to a bank
Impacted by Bank Activity	✓Yes ✓Counterparties will often adjust intraday credit extensions to reflect business activity from other areas of
Ability to forecast	✓Some intraday credit facilities are disclosed and well known to a bank. ✓Others are not disclosed but can often be inferred from historical data

Description	✓Cash, money market deposits, and short-term government debt (e.g., Bills) which can be quickl
Impacted by Client Activity	✓Yes ✓Client may be sources of liquidity in converting liquid assets to cash (e.g., repo transact
Impacted by Bank Activity	✓Yes ✓Bank trading and investment portfolio activity can impact the amount of liquid assets ava
Ability to forecast	✓Liquid assets held in the investment portfolio and in other money market investments tend to have low v

Description	✓Fed Funds ✓Eurodollar borrowing ✓Overnight deposits
Impacted by Client Activity	✓Yes ✓Clients may be direct sources of liquidity for overnight borrowing (e.g.,. overnight deposit)
Impacted by Bank Activity	✓No ✓Other bank lines of business do not regularly supply overnight funding
Ability to forecast	✓Client supply of overnight borrowing tends to be fairly predictable, but with moderate volatility

Description	✓Other term deposits ✓Repos from FHLB ✓Money market funds, etc.
Impacted by Client Activity	✓Yes ✓Clients may be direct sources of liquidity for term funding
Impacted by Bank Activity	✓No ✓Other bank lines of business do not regularly supply term funding
Ability to forecast	✓Client supply of term funding tends to be predictable, but with low volatility

4 Governance of Intraday LRM

- Active risk management
  - Classify settlement and systemic risks as components of risk taxonomy, and critically, incorporate these risks into the firm's risk appetite framework.
- Integration with risk governance: Oversight of intraday liquidity risk management is integrated into the bank's overall risk oversight structure. Three lines of defense model:
  - Treasury: manages the intraday funding positions of risk management programs related to funding activities.
  - Corporate Risk Management: oversees funding-related policy and procedures.
  - Internal Audit: independently assesses the bank's adherence to intraday risk policies and procedures.
- Risk assessment
  - Through risk self-assessments, settlement risks related to existing and potential new products and operational processes are identified, measured, and evaluated.
- Risk measurement and monitoring
  - The amount of intraday credit the institution is extending to clients.
  - The amount of intraday credit the institution utilizes.

5 Measures for understanding intraday flows

- Total payments
  - A bank and its intraday risk management should maintain statistic concerning the amount of payments it makes on all electronic payments systems in which it participants.
  - For example:
    - ∗ Total payments sent and received for non-financial institution or financial institution clients/ bank activity.
    - ∗ Net position in the settlement account at any time of day in aggregate.
- Other cash transactions
  - A bank should also track its intraday and end-of-day settlement positions at all financial market utilities in which it participates, such as collateral position in SSN.
- Settlement positions
  - If complete data to reconstruct account positions at any time of day is not available, at a minimum a bank should maintain data on its settlement positions with all its FMUs.
- Time sensitive obligations
  - For those transactions require completion at a specific time during the day, such as return of borrowing, margin payments. Failure to settle may result in a financial penalty.

- Total intraday credit lines to clients and counterparties
  - Bank should have data regarding average and peak usage, and the ability to model activity at the client and portfolio levels.
- Total bank intraday credit lines available and usage
  - The amount of intraday credit that a bank relies on in business-as-usual conditions and the maximum amount of intraday borrowing it can draw down.
  - Captures the amount of committed and uncommitted intraday credit (and usage thereof) the bank has at its disposal, ideally across all its cash and settlement accounts.

6 Measures for quantifying and monitoring risk levels

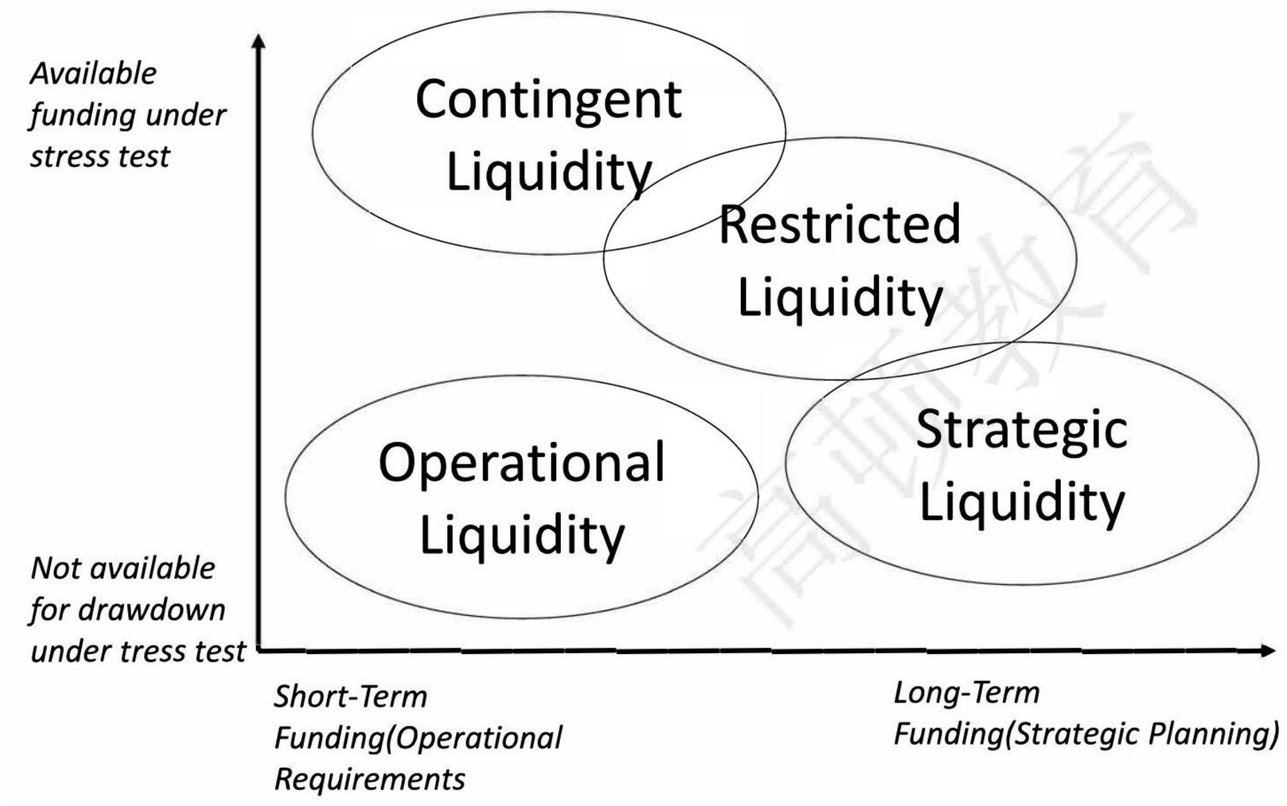
- Daily maximum intraday liquidity usage
  - Formula: $\frac{\text{the day's largest negative balance}}{\text{the size of the committed or uncommitted credit line}}$
    - ∗ This calculation does not need real-time monitoring of an account to capture all of the negative positions.
  - Typically, the peak and average of this metric is tracked over a period of time (e.g. monthly).
- Intraday credit relative to tier 1 capital
  - Formula: $\frac{\text{Intraday credit}}{\text{Tier 1 capital}}$
    - ∗ The measure should be tracked for total intraday credit and unsecured intraday credit, available and used.
    - ∗ Represents the intraday settlement risk.
  - Available unsecured intraday credit relative to an institution's tier 1 capital is a rough measure of the inadvertent systemic risk that the institution poses to the financial system.
- Client intraday credit usage
  - Formula: $\frac{\text{Client's peak daily intraday overdraft}}{\text{established (committed or uncommitted) credit line}}$
  - Provides useful insights for understanding how client activity impacts the bank's ability to manage its own intraday liquidity
- Payment throughput
  - These measures track the percentage of outgoing payment activity relative to time of day.
  - Help the bank identify and monitor its peak periods over time and the correlation of this activity with its intraday liquidity on hand and intraday credit usage.

考点 3 liquidity Stress Testing and Reporting Ch9

1 Four categories of funding liquidity

- Operational liquidity: The cash that is needed to fund the business daily.
  - May be quite volatile
  - May need cushion for unpredictability of daily settlement
- Restricted liquidity: Liquid assets only for specifically defined purpose, such as collateralization.
  - Unavailable to meet general financial obligations under a liquidity stress test.
- Strategic liquidity: The cash that is held by the institution to meet future business needs, outside the course of normal operations.
  - Fixed asset purchases or M&A
- Contingent liquidity: Liquid asset buffer composed of high quality and liquid financial assets.

– Meet general financial obligations under a stress scenario.



2 Estimating stressed liquidity asset buffer

- Stressed liquidity asset buffer: indicates the adequacy of the current liquid asset buffer given stress scenario assumptions.
- Calculation formular:
  – stressed liquidity asset buffer = normal liquidity asset buffer - stressed cash outflow + stressed cash inflow
    * Normal liquidity asset buffer: the amount of contingent liquidity currently held.
    * Stressed outflow: require early settlement of liabilities or the inability to rollover debt.
    * Stressed inflows: such as drawdowns on liquidity facilities available to the institution.
- Example
  – Bank ABC has 1 million dollars in highly liquid asset. In its stressed scenario assumption, the retail customer will withdraw $200,000 deposit and the debt cannot roll over is $600,000. The bank can use the remaining $500,000 of its available liquidity facility. What is the estimate stressed liquid asset buffer?
  – Answer:
    * 1,000,000-(200,000+600,000)+500,000=700,000 dollars

3 Design of the model

- Organizational scope
- Scenario development
- Development of assumptions
- Outputs of the model
- Governance and controls
- Integrations with other risk models

4 Organizational scope

- Liquidity transfer restrictions
  - Liquidity may be trapped in certain legal entities and cannot be transferred freely throughout the organization
- Currency
  - Perform liquidity stress test in the currency of the entity being tested, and take liquidity impact of currency conversion requirements for consideration
- Regulatory jurisdiction
  - Perform separate stress tests for the foreign entities

5 Scenario development

- Historical scenarios
  - Historical scenarios are based on actual liquidity failures and attempt to translate those events to the financial institution performing the stress test.
  - Disadvantage:
    * The existence of few examples with limited data
    * The past is not always indicative of the future
  - Forward-looking view in which the financial institution experiences severe liquidity stress:
    * Distinguish between systemic and idiosyncratic risk
    * Distinguish between levels of severity.
    * Clearly define the scenarios.
  - Reverse liquidity stress tests: supplement to hypothetical scenanos.

6 Development of Assumptions

- Develops liquidity stress testing assumptions the institution should do the following:
  - Qualitatively assess the expected liquidity behavior for each type of cash flow to determine where there is significant liquidity risk.
  - Determine the appropriate level of segmentation for each type of risk based on an assessment of behavioral differences, bearing in mind any limitations in ongoing data availability.
  - Qualitatively assess and order by rank varying levels of liquidity risk for each segmentation factor, potentially utilizing a scoring system.
  - Develop quantitative modeling assumptions based on any historical data available, such as experiences during the financial crisis, or available from other sources such as peer benchmarking.
  - Develop matrices of relative modeling assumptions based on scored risk levels and baseline historical data.
  - Adjust assumption matrices as appropriate for each stress scenario, for example, reflecting differences in relative overall severity or assumptions concerning idiosyncratic or systemic risk.
- The following assumptions can have an outsized impact on the results of the stress test, and should be considered carefully in developing the model:
  - Investment portfolio haircuts
  - Deposit outflows
  - Unsecured wholesale funding
  - Collateral requirements

  - Other contingent liabilities
  - Business dial back

7 Outputs of the model

- Stress testing assumptions
  - Overall stress level by the scenarios
  - Indication of whether the scenario is systemic, idiosyncratic, or both
  - Contribution of economic, market, and firm-specific events
  - Cash flows resulting from the scenario
- Liquidity position metrics
  - Main objective is to determine how much liquidity is available compared to the net cash outflows.
- Prospective liquidity position metrics
  - Highlight any times during the stress horizon where survival would require significant transactions such as capital infusion and debt issuances.
- Capital and performance metrics
  - Captures the impact of any capital actions required to support liquidity during the stress period.

8 Governance and controls

- Asset-liability committee(ALCO)
  - The ALCO is in charge of the liquidity stress testing framew ork.
  - Ensuring the establishment, review, and approval of a liquidity stress testing policy.
  - Suggesting and approving liquidity risk scenarios.
  - Setting liquidity risk policy limits dependent on stress test outcomes.
- Treasury unit (the first line of defense)
  - Maintenance of liquidity stress testing procedures.
  - Recommending stress test scenarios.
  - Analyzing liquidity characteristics of the assets and liabilities.
  - Recommend assumptions used in stress testing to ALCO.
  - Reports results of stress testing.
- Risk management department (the second line of defense)
  - Administering the liquidity risk stress testing policy
  - Reviewing and providing effective challenge of the scenario design and assumptions
  - Ensuring the institution's approach to liquidity stress testing is in line with acceptable industry practices and regulatory rules and guidance
  - Reviewing and approving the liquidity stress test-based limits
  - Monitoring of liquidity stress test-based limits
  - Ensuring the institution's ALCO, executive management, and board are kept well-informed of the bank's liquidity risk profile as indicated by the stress testing results
- Internal audit (the third line of defense)
  - Regular checks on the stress testing process, procedures, and controls as a confirmation.

- Model risk management
  - A separate group that validates the stress testing model to ensure it is consistent with the firm's model risk management policy.

9 Integrations with other risk models

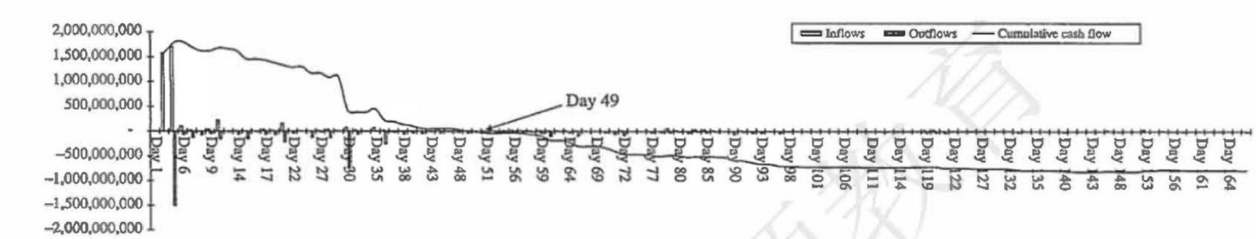
- Liquidity stress testing and capital stress testing:
  - Liquidity stress test incorporates any required capital infusions of subsidiary entities. Capital stress testing should consider liquidity stress evaluation.
- Liquidity stress testing and asset liability management:
  - Liquidity stress test scenario framework, and liquidity risk EWIs should not neglect to include the possibility of an interest rate shock and the potential impact such an event would have on indeterminate liabilities.
- Liquidity stress testing and funds transfer pricing:
  - Cost of carrying cash buffer required in liquidity stress testing will be priced through FTP framework.

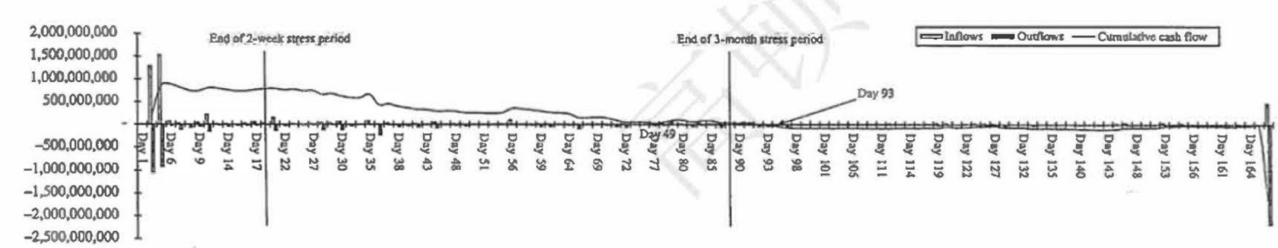
10 Liquidity Stress test report

- Definition: Stress test output results (report)
  - Help senior management to understand the liquidity position of the bank, enabling them to take mitigating action if deemed necessary.
    - ∗ Cash flow survival report
    - ∗ Stressed cumulative cash flow forecast
    - ∗ Line-by-line stress test result report

11 Cash flow survival report

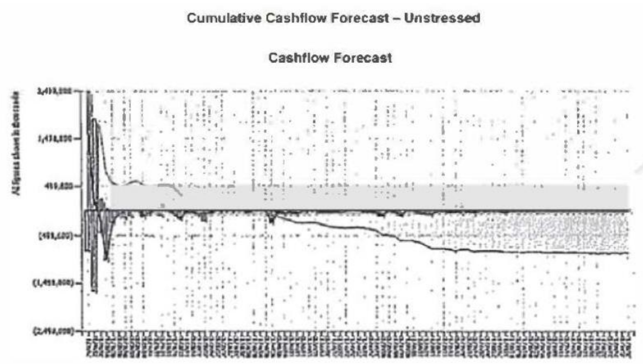
- The primary stress test output:
  - A report under business as usual (BAU) conditions and one after taking the mitigating actions (such as liquidating securities, and accessing contingency funding sources).
  - Frequency: weekly for firm-specific/daily for market-wide.

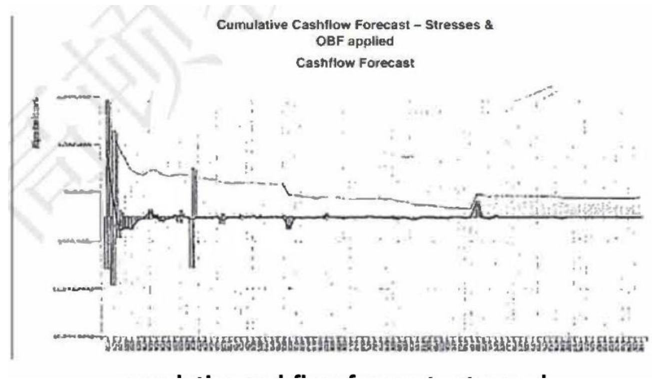




12 Stressed cumulative cash flow forecast

- The second stress test output shows the stressed cumulative cash flow forecast taking into account the immediate sale repo of marketable securities.





13 Line-by-line stress test result report

- Show the results of individual shocks on the liquidity ratio, and the probability of each result occurring and is a part of routine stress testing undertaken by Treasury or Risk Management.
  - Individual shocks like:
    * Reduction in liquid assets
    * Decrease in liabilities
    * FX mismatch
  - Frequency: quarterly or required by the regulator.
- Reduction in liquid assets

Reduction in Liquid Assets	Severity	Description	Sight − 8 Day	Sight − 1 Month	Probability	Impact
Change in repo criteria	Light	Rating category 1 notch downgrade	8.46%	1.35%	50%	30
	Moderate	Rating category 2 notch downgrade	2.34%	0.12%	20%	70
	Severe	Rating category 3 notch downgrade	-15.2%	-18.2%	1%	90
Mark-to-market reduction in value of assets	Light		8.46%	1.35%	60%	20
	Moderate		2.34%	0.12%	40%	30
	Severe		-15.2%	-18.2%	5%	70
Increased haircut on assets	Light		8.46%	1.35%	70%	25
	Moderate		2.34%	0.12%	30%	45
	Severe		-15.2%	-18.2%	8%	80
Unavailability of repo facilities	Severe	Treat all marketable securities as illiquid (i.e., allocate to final legal maturity time buckets)	-15.2%	-18.2%	5%	100

table 9: Stress Tests −Individual Shocks

考点 4 Liquidity Risk Reporting Ch10

1 Liquidity Risk Reporting

- A bank will produce a number of liquidity reports in the normal course of business, on a daily, weekly, monthly and quarterly basis.
  - The main liquidity reports and its frequency are required by the regulator.

  - ALCO view regulator's requirements as a minimum requirement, and supplement it with additional management information as desired.

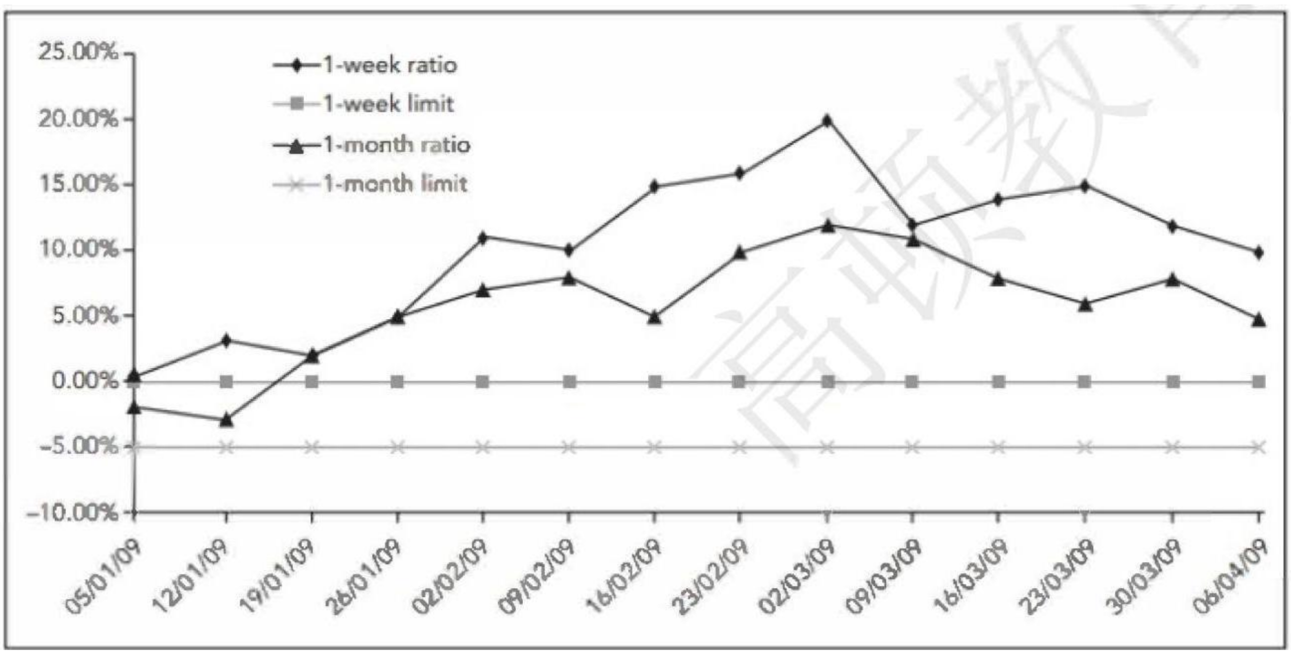
- Reports are an important part of liquidity management information (MI).
- Summary and Qualitative report
  - Summary report: Liquidity report MI for senior management should be presented as a 1-page summary of the key liquidity metrics.
    * Increase the chance that the report will be read and noted at senior management level
  - Summary report example-part 1

table 10: Monthly Liquidity Snapshot - Cumulative Liquidity Report ('000)

	1 Day	2 Day	1 Week	2 Week	1 Month	2 Months	3 Months	6 Months	1 Year
Cumulative Net Cash Balance	80	80	–1,260	–1,260	–1,180	–1,408	–1,150	–850	–850
Other Forecast Inflows									
Other Forecast Outflows									
Cumulative Cash Gap	80	80	–1,260	–1,260	–1,180	–1,408	–1,150	–850	–850
Counterbalancing Capacity	180	180	635	640	748	748	1,238	1,238	1,238
Liquidity Gap	260	260	–625	–620	–432	–660	88	388	388
Limit									
Variance	260	260	–625	–620	–432	–660	88	388	388

∗ * Cash gap turns negative between 2 - day and 1 - week
∗ * Liquidity gap turns negative between 2 - days and 1 - week

  - Summary report example-part 2



  - Summary report example-part 3

	Current month	Previous month	Change
LIQUIDITY RISK FACTOR	22.2	25.1	↓
LOAN-TO-DEPOSIT RATIO	96%	93%	↑
NET INTER-GROUP LENDING	(1,228)	(1,552)	↓

  – Qualitative report: should be completed for Head Office Group Treasury on a monthly basis, for banking groups that operate across country jurisdictions and multiple subsidiaries.
    * Assist the group to better understand the liquidity position in each country.

- Qualitative report-Example

GROUP TREASURY QUALITATIVE REPORTING

Monthly Liquidity Highlights

Template for Branches and Subsidiaries to Summarise Main Liquidity Changes for the Last Month

To be Provided a Fixed Specified Date Each Week

Points to consider:

1. Explain significant changes in your 1-week and 1-month liquidity ratios
2. Explain any changes to your cash and liquidity gap in your Cumulative Liquidity model
3. Explain significant changes to the Liquidity Risk Factor
4. Explain growth or shrinkage of asset books
5. Detail any changes to inter-group borrowing/lending position; detail the counterparties for any large-size deals
6. Any increase/decrease in corporate deposits, detail large dated transactions with an estimated confidence level of roll-over
7. Any increase/decrease in retail deposits
8. Average daily opening cash position

This list is not exhaustive and any other relevant points are welcome.

2 Benchmark liquidity reports for regulated banks in the UK

- Deposit tracker report
- Daily liquidity report
- Funding maturity gap (mismatch) report
- Funding concentration report
- Undrawn commitment report
- Liability profile
- Wholesale pricing and volume

3 Deposit tracker report

- Definition: A simple report of the current size of deposits by customer type and forecast of what the level of deposits are expected to be going forward. LTD ratio is also provided.
- Identified liquidity position: Loan to deposit ratio (LTD)
  – Too high: may not have enough liquidity to cover any unforeseen fund requirements.
  – Too low: may not be earning as much as it could be.
- Frequency: weekly and monthly

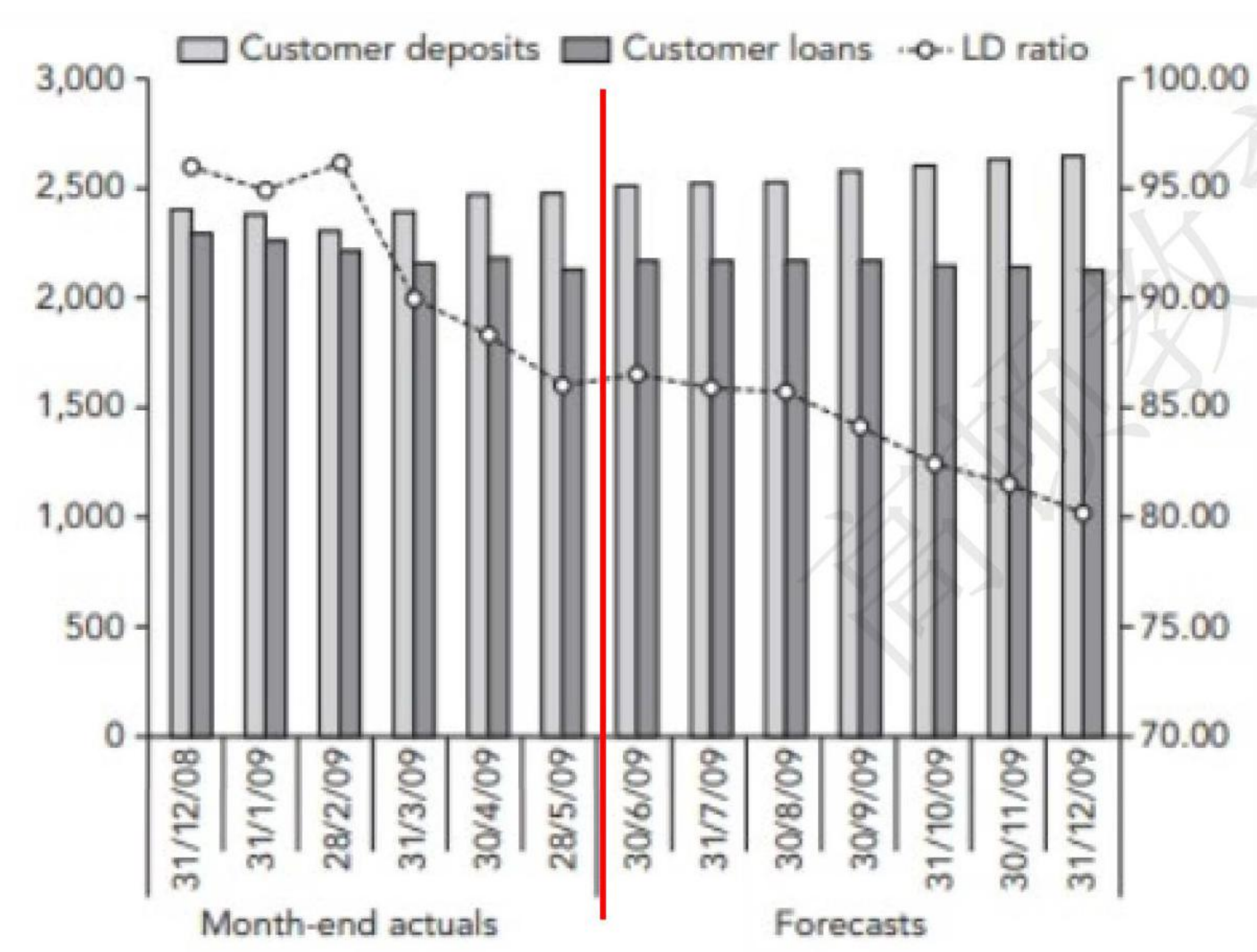
current　　　　　　forecast

Customer deposits

Customer loans

Deposit Tracker	Month End Actuals						Forecasts						
	31/12/2008	31/01/2009	28/02/2009	31/03/2009	30/04/2009	28/05/2009	30/06/2009	31/07/2009	30/08/2009	30/09/2009	31/10/2009	30/11/2009	31/12/2009
Eligible Correspondent Banks	482,236	431,166	485,302	507,193	536,907	493,930	515,753	520,753	520,753	520,753	525,753	530,753	535,753
Corporate Client Deposits	449,871	375,849	248,677	263,267	243,710	280,248	273,893	273,893	273,893	273,893	273,893	273,893	273,893
Private Bank Client Deposits	14,168	23,334	18,990	102,174	102,582	99,119	99,123	99,123	99,123	124,123	124,123	124,123	124,123
Local Authority Deposits	196,624	195,814	192,100	226,622	267,001	325,016	333,287	343,287	348,287	373,287	393,287	413,287	433,287
Retail Bank Deposits	1,234,799	1,318,219	1,323,738	1,264,323	1,293,918	1,258,133	1,264,025	1,264,025	1,264,025	1,264,025	1,264,025	1,264,025	1,264,025
Eligible Private Bank Correspondent Banks	24,864	37,456	38,358	37,196	37,388	35,512	35,529	35,529	35,529	35,529	35,529	35,529	35,529
Treasury Sales	5,775	5,198	4,477	3,846	822	770	763	763	763	763	763	763	763
Total Customer Deposits:	2,408,337	2,387,036	2,311,642	2,404,621	2,482,328	2,492,728	2,522,373	2,537,373	2,542,373	2,592,373	2,617,373	2,642,373	2,667,373
M/M +/–		–21,301	–75,394	92,979	77,707	10,400	29,645	15,000	5,000	50,000	25,000	25,000	25,000
W/W +/–													
Mend +/–													
Drawdown-Month ahead							76,374	8,526	4,479	5,298	2,911	3,136	5,570
Repayment-From loans schedules							37,888	11,920	4,429	2,326	25,800	7,965	20,863
Forecast Monthly loan +/–							38,486	–3,394	50	2,972	–22,889	–4,829	–15,293
Total Customer Loans:	2,305,766	2,266,004	2,223,145	2,166,076	2,194,016	2,145,648	2,184,134	2,180,740	2,180,790	2,183,762	2,160,873	2,156,044	2,140,751
Loan-to-deposit %	95.74	94.93	96.17	90.08	88.39	86.08	86.59	85.94	85.78	84.24	82.56	81.59	80.26

LTD ratio



4 Daily liquidity report

- Definition: A straightforward spreadsheet detailing the bank's liquid and marketable assets and liabilities, up to 1-year maturity. Each branch and subsidiary will complete this report.

- Input data: Summarizes the bank's liquid assets, liabilities by maturity/tenor.
- Output metrics: Provides an end-of-day of the bank's liquidity position like liquidity gap, cumulative cash gap.

• Frequency: daily

• Input data

ASSETS	1 Day	2 Days	1 Week	2 Weeks	1 Month	2 Months	3 Months	6 Months	<1 Year	>1 Year	Total	
Non-marketable Securities & CDs	0	0	0	0	0	0	0	0	0	0	0	
Retail call a/c	6,754	0	0	0	0	0	0	0	0	0	6,754	
Retail time deposit	432	3,533	0	4	4	4	4,529	9	1,619	420	23,749	34,299
Inter-Group call	4,725	0	0	0	0	0	0	0	0	0	4,725	
Inter-Group time	19,199	89,848	281,434	90,150	135,378	73,895	60,768	21,379	5,912	244	778,207	
Other bank call	56,568	0	0	0	0	0	0	0	0	0	56,568	
Other bank time	5,465	148,347	188,620	7,833	89,500	30,020	0	25,319	128	83,985	579,217	
Corporate call	30,658	0	0	0	0	0	0	0	0	0	30,658	
Corporate time	13,936	112,975	35,335	53,432	10,923	159,894	133,718	63,182	116,551	1,365,637	2,065,583	
Government bonds	161,011	0	19,361	35	335	303	9	868	678	84,857	267,457	
Total Assets	298,748	354,703	524,750	151,454	236,140	268,641	194,504	112,367	123,689	1,558,472	3,823,468	
Average remaining duration of assets			*44.59 Months*									

Liabilities	**1 Day**	**2 Days**	**1 Week**	**2 Weeks**	**1 Month**	**2 Months**	**3 Months**	**6 M**
Retail call	97,482	0	0	0	0	0	0	
Retail time	19,780	18,250	120,492	65,241	123,345	113,741	157,361	12
Inter-Group call	50,996	0	0	0	0	0	0	
Inter-Group time	72,276	136,744	351,130	233,319	60,987	161,120	43,881	10
Other bank call	235,097	0	0	0	0	0	0	
Other bank time	16,088	116,319	135,796	21,210	121,181	85,287	192,512	6
Corporate call	60,085	0	0	0	0	0	0	
Corporate time	17,937	94,226	173,805	152,144	94,976	111,413	47,748	1
Government	3,689	6,005	58,163	109,046	37,671	204,339	54,477	3
Total Liabilities	573,430	371,544	839,386	580,960	438,160	675,900	495,979	34
Undrawn commitments								
Average remaining duration of liabilities								

• Output metrics

table 11: Cumulative Liquidity Report

	1 Day	2 Day	1 Week	2 Week	1 Month	2 Months	3 Months	6 Months	1 Year
Cumulative net cash balance	-111734	-104480	-301843	-642230	-738261	-1040588	-1884251	-2122457	-2099158
Other forecast inflows									
Other forecast outflows									
Cumulative cash gap	-111734	-104480	-301843	-642230	-738261	-1040588	-1884251	-2122457	-2099158
Counterbalancing capacity	8676	8676	797674	804186	828686	877686	979996	1092696	1092696
Liquidity gap	-103058	-95804	495831	161957	90426	-162902	-904256	-1029761	-1006462
Limit									
Variance	-103058	-95804	495831	161957	90426	-162902	-904256	-1029761	-1006462

5 Funding maturity gap (mismatch) report

• Definition: shows the maturity gap (maturity mismatch) per time bucket, for all assets and liabilities, with an adjustment for liquid securities.

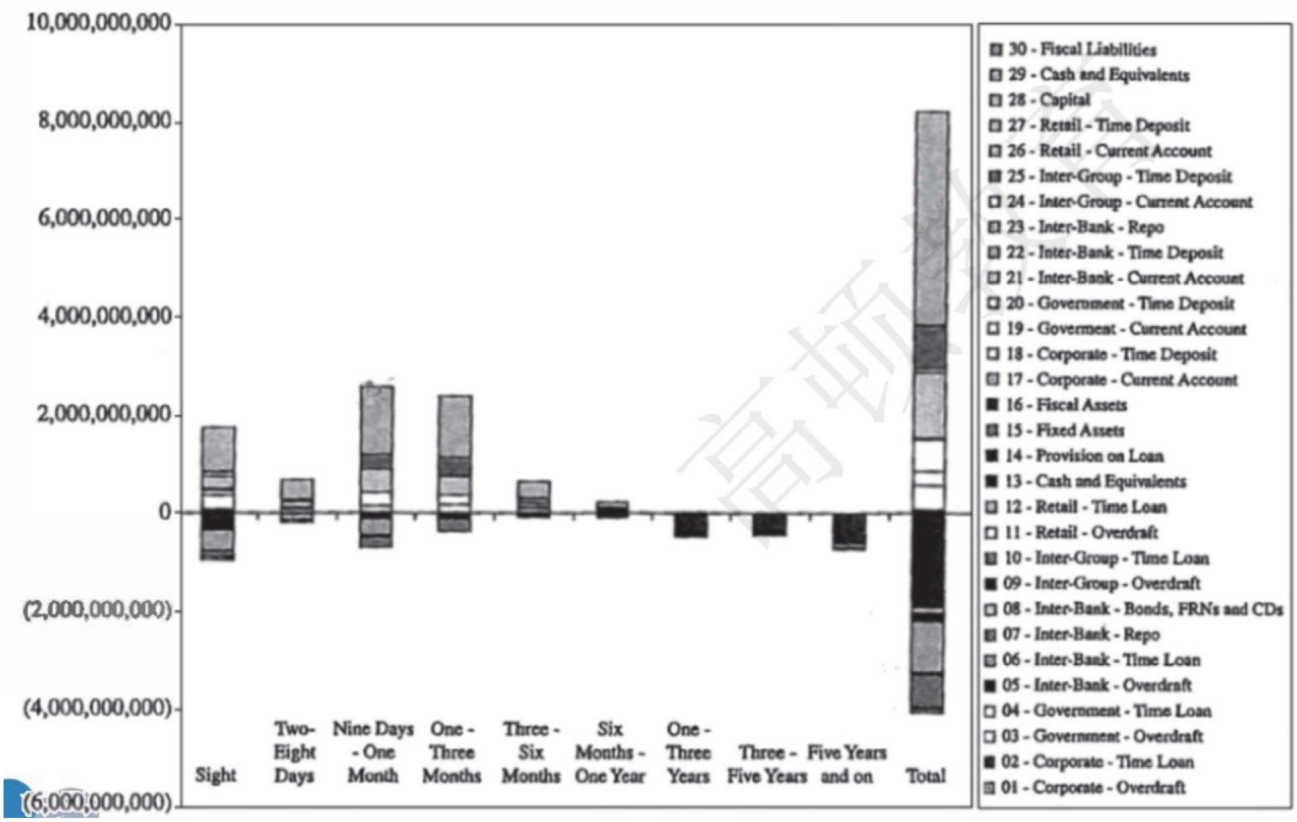
- Identified liquidity position: cumulative liquidity cash flow of daily liquidity report is included in this report.

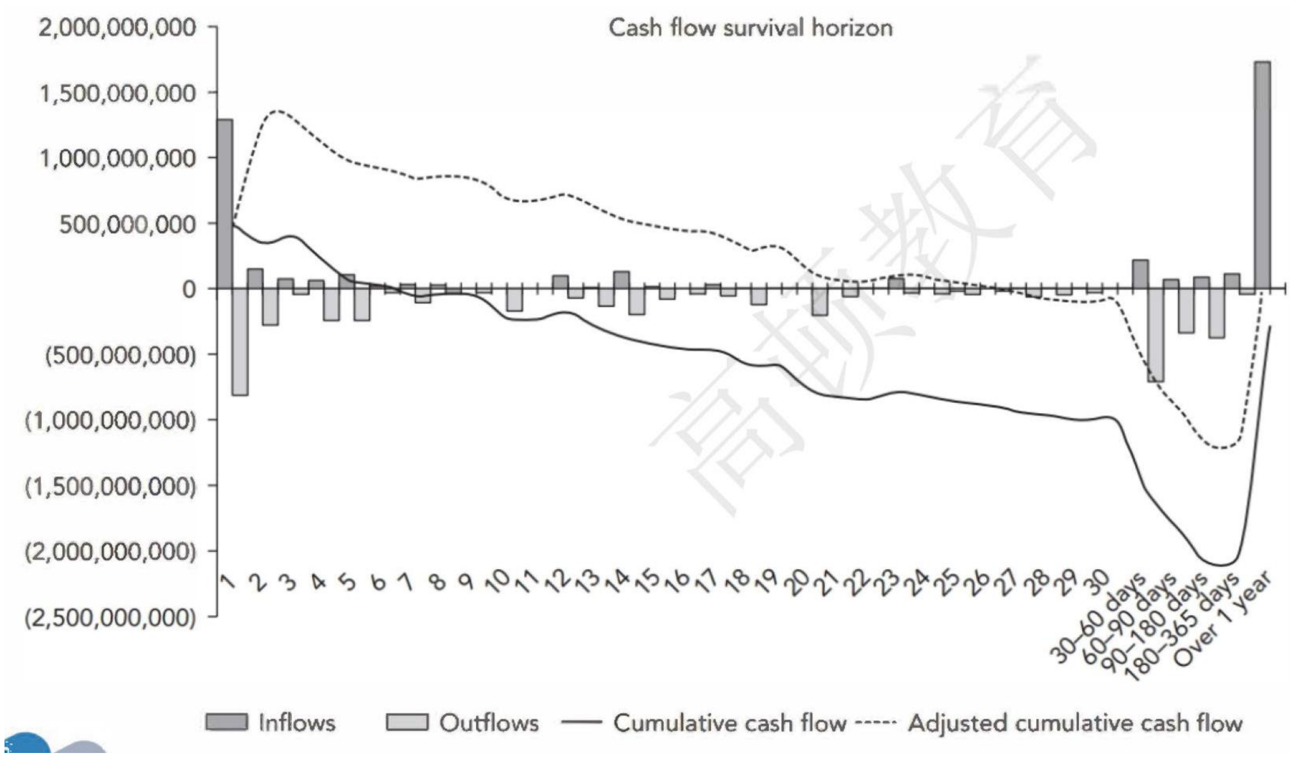
table 12: Liquidity Metrics

		Limit
1-week Ratio	11.13%	0.00% OK
1-month Ratio	2.03%	–5% OK
Liquidity Risk Factor	3.85	

- 0-days survival horizon: required by Basel III.

• Frequency: monthly



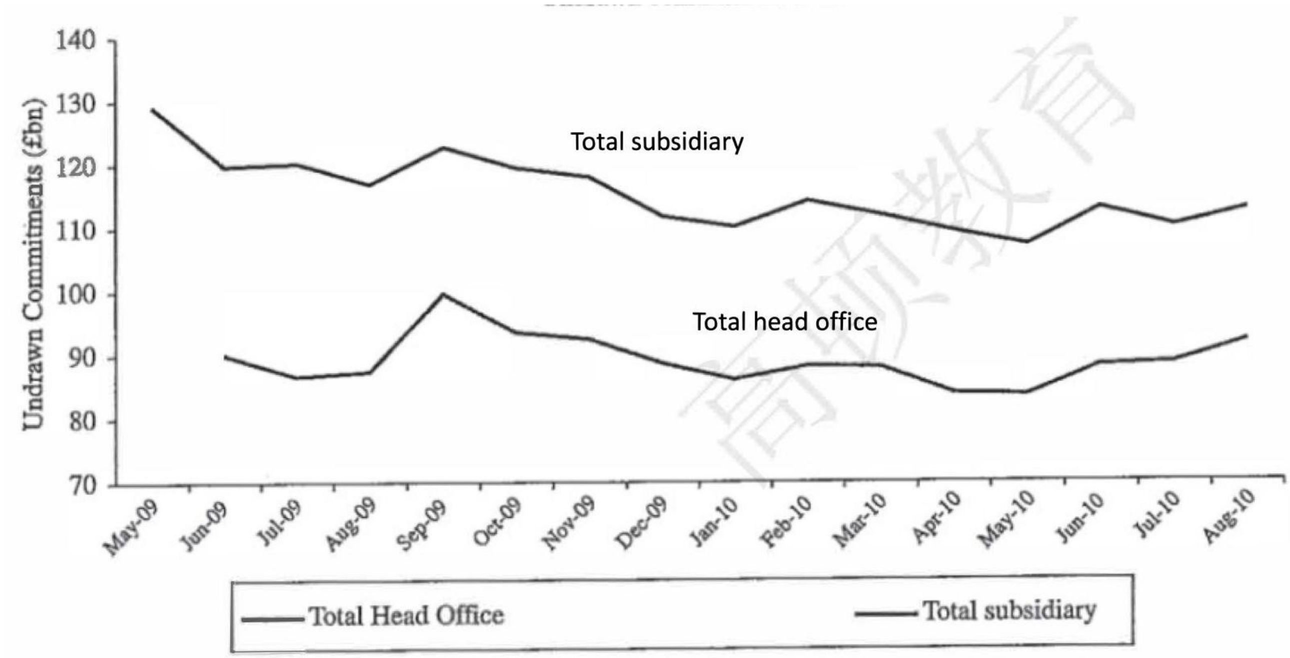


6 Funding Concentration Report

- Definition: Funding source concentration reports
  - Funding diversity: A bank should not become over-reliant on a single source, or sector, of funds. A deposit of 5% of total liabilities should be treated as large by ALCO.
    - The way to increase diversity: Increase its liabilities rather than asking the depositor to remove some of the fund to avoid damaging the relationship.
  - Frequency: monthly

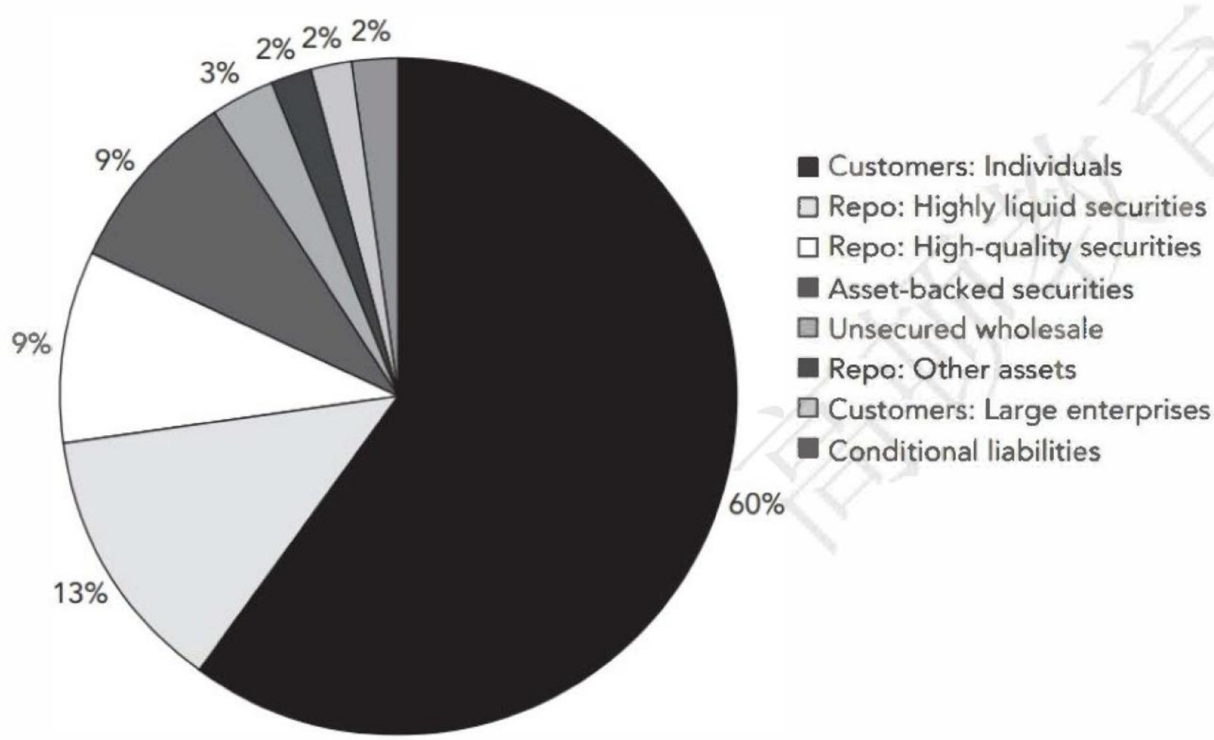
7 Undrawn Commitment Report

- Definition: shows the existence of undrawn commitment trend over time.
  - Off balance sheet products: potential stress points for a bank's funding requirement and exacerbate funding shortages at the wrong time.
    - Liquidity lines, revolving credit facilities, letters of credit, guarantees
- Frequency: monthly



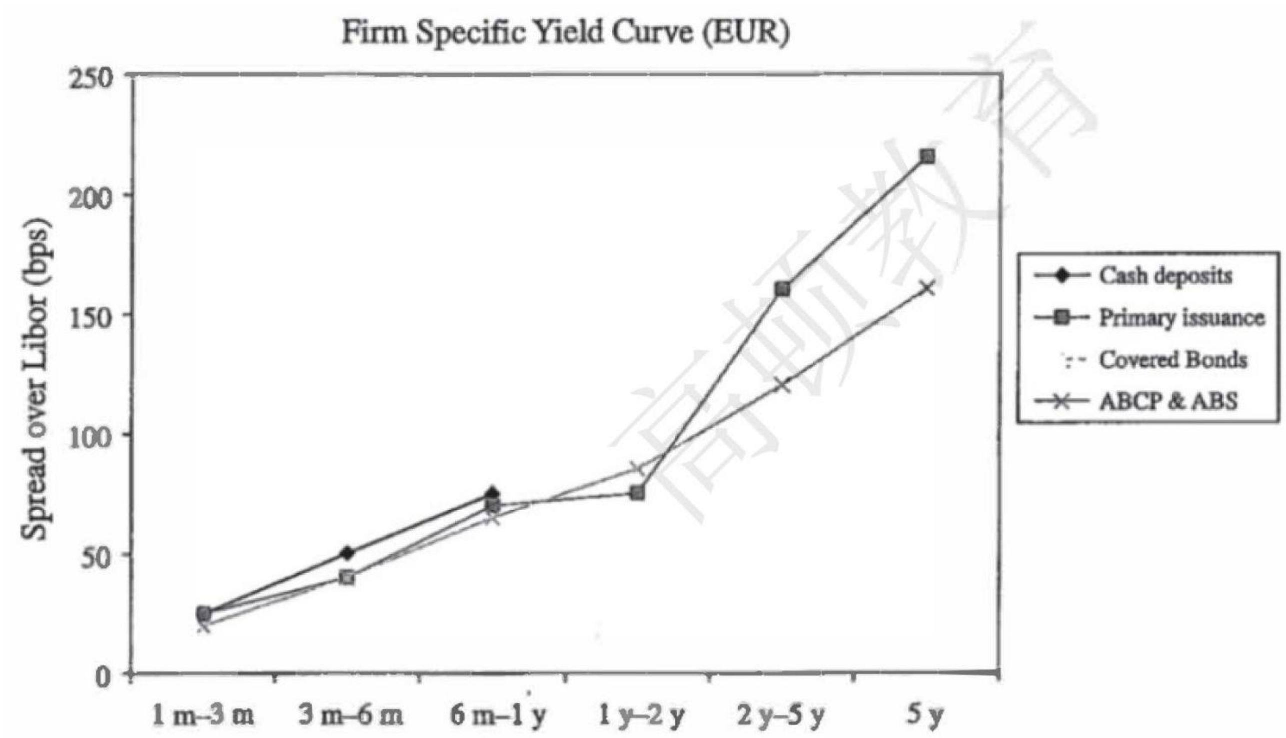
8 Liability Profile

- Definition: A simple breakdown of the share of each type of liability at the bank. The report format can be set to the user's desired choice.
  - The total liabilities are made of the following categories:
    * Customers: individual and large enterprises
    * Repo
    * Asset-backed securities
    * Unsecured wholesale, etc.
- Frequency: monthly



9 Wholesale Pricing and Volume

- Definition: a report of firm-specific yield curve.
  - Liquidity strength can be gleaned from looking at a bank's funding costs, and the composition of its funding by product.
  - An early warning of a particular bank experiencing funding stress if its funding yield curve is rising materially above that of its peer group.
- Frequency: at least quarterly



考点 5　Contingency Funding Planning Ch11

1 Introduction

- Definition: Contingency Funding Plan (CFP) provides a structured approach for developing and implementing the institution's financial and operational strategies for effectively managing such contingent liquidity events during periods of severe market and financial stress.
  - Business as usual funding (BAU) : low-impact, highprobability event.
  - Contingency Funding Plan （CFP）: high-impact, lowprobability events.

2 Relationship with liquidity stress test

- Link the stress test results and other related information as inputs to the CFP governance.
- Liquidity stress testing is helpful to establish limits and escalation levels.

3 Key design considerations

- Aligned to business and risk profiles
  - Aligning the institutions' risk appetite to the CFP framework, through quantifiable EWIs, limits, and escalation levels.

  - Refresh CFP as the institution's business and risk profile changes over time. Leading institutions develop the CFP to be more forward looking and flexible.
    - Corporate and business strategies change
    - New products and services
    - Macroeconomic and market environments

- Integrated with broader risk management frameworks
  - An integrated part of the institution's liquidity risk management and firm-wide risk management frameworks:
    - Enterprise risk management (ERM)
    - Capital management
    - Business continuity
    - Crisis management

- Operational and actionable, but flexible playbook
  - A menu of possible contingency actions that management can undertake in different stress scenarios and at graduated levels of severity.
  - As a playbook, the effectiveness of the CFP lies in its operational readiness.

- Inclusive of appropriate stakeholder groups
  - Involvement of appropriate stakeholder groups:
    - Asset liability committee (ALCO)
    - Risk and capital committee
    - Investment committee
    - Business units:
      - Finance, corporate treasury, risk, operations , technology
  - Ensure internal and external stakeholders to remain confident in the institution's financial strength.

- Supported by a communication plan
  - In any crisis, the coordinated and timely communication of information to stakeholders is critical.
  - External communication to clients, analysts, counterparties, and regulators with timely and accurate information is critical.
    - Help to reinforce confidence in the institution and mitigate potential risk that rumors, and fears do not further precipitate and adversely impact the institution.

4 Key components of a CFP framework

- Governance and oversight

Roles	Responsibilities
Corporate treasury	✓In consultation with the CFO and others, may invoke CFP and convene Liquidity Crisis Team
Liquidity crisis team(LCT)	✓Designing the CFP ✓Providing recommendations on CFP actions ✓Central point of contact, ensure effective coordination across the organization as well as with external st
Management committee	✓Oversight of LCT ✓Reviewing specific recommendations for and coordination of CFP actions
Board of directors	✓An advisor and counsel to LCT and management ✓Evaluating CFP actions

- Scenarios and liquidity gap analysis
  - Align the CFP stress scenarios to liquidity stress testing framework, as well as to other frameworks such as the recovery and resolution plans.
  - CFP may contain additional liquidity-related stress scenanos.
    * lntraday debit cap with Fedwire is exceeded
    * Specific counterparties fail
    * Federal Home Loan Bank funding becomes unavailable
- Contingent actions: help strengthen the institution's liquidity position.
  - Reducing the amount of lending activity
  - Offering higher rates on deposits
  - Choosing not to reinvest securities when they mature
  - Increasing securitization activities
  - Disposing of liquid assets
  - Issuing subordinated debt
  - Disposing of business unit
- Monitoring and escalation
  - Early warning indicators
  - Market and business factors
    * Spike in market volatility (e.g., VIX)
  - Liquidity health measures
    * Projected net funding requirements current unused funding capacity
    * Non-core funding to long-term assets
    * Overnight borrowings to total assets
    * Short-term liabilities to total assets
    * Funding sources concentration
    * Funding maturity profile
    * Used capacity to total borrowing capacity
    * Liquid assets to volatile liabilities
    * Unpledged eligible collateral to total assets
    * Loans to commitments
  - Escalation Levels: 3 to 5 escalation levels
    * Level 1: Triggered by stress test results indicating a greater decline in liquidity than the institution's risk appetite targets and/or a shift in the market's perception of the institution.
    * Level 2: Triggered when institution experiences noticeable markets and/or idiosyncratic events that are adversely impacting its business and liquidity risk profile.
      · Focus on immediate and short-term horizon.
    * Level 3: Triggered when dramatic steps should be taken to stabilize the liquidity position.
      · Focus on survival.
- Data and reporting
  - Reports capture intraday liquidity positions, track exposure to contingent liabilities, and monitor capacity usage in funding sources.
  - Reports convey methods used to determine liquidity coverage for upcoming liabilities and funding.

- Increasing the frequency of liquidity management reporting benefit the institution.